INTERNTRACK: INTERNSHIP MANAGEMENT SYSTEM FOR UNIVERSITY OF CABUYAO

A Capstone Project Proposal

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CHAPTER I
THE PROBLEM AND ITS SETTING

Introduction

The University of Cabuyao, also referred as Pamantasan ng Cabuyao, is significant in offering affordable higher education to the students within the City of Cabuyao, Laguna and its surrounding communities. With the institution having begun its operation on April 16, 2003, there has been an increase in the academic offerings, as well as the intensification of programs that facilitates not only academic success, but also in the preparation of professionals. In this regard, the university encourages learning experiences that enable students to utilize the classroom knowledge to real workplace environments.

The internship or on-the-job training (OJT) program is one of the greatest of such learning experiences. Internships can be considered an essential role in experiential learning since they allow students to translate theoretical knowledge to actual work settings, acquire professional skills, and become more job-ready [1]–[4]. Students are introduced to the organizational processes, performance expectations and expectations at work place when participating in internship activities, polishing the organization technical skills, communication skills and flexibility. Through this, the internship is effective in closing the connection between education and professional practice.

In the University of Cabuyao, most of the operations of the internship activities are still quite manual and fragmented. Student profiling, validation of attendance, journal submission, document compliance, supervision and performance evaluation are usually processed individually as per individual files, reiterated follow-ups, slow verification processes and so on. Consequently, documentation becomes challenging and records are hard to bundle, monitoring is not ever time-based and stakeholders have a poor idea on how the students are faring and adhering.

As a measure to address such constraints, schools and colleges have progressively welcomed web-based applications, mobile-responsive sites, and storage of documents digitally, as well as centralized data operations in order to enhance efficiency, accessibility and real-time tracking. These technologies facilitate quicker connectivity, better records maintenance and uniform decision-making. Under internship administration, these systems may be used to enhance the coordination of stakeholders, automate the submission and validation of requirements and make data on student progress readily available.

Despite the earlier research studies demonstrating the capabilities of the internship management systems to enhance monitoring, document handling and performance evaluation, most of the latter were designed to operate in the institutional context, where workflows and policies as well as operational structures vary. According to previous research [18], [19], [21], [27], [30] the success of an internship platform relies on the way the platform fits in the local processes, user functions, and monitoring need. Furthermore, there are less works that focus on localized workflow structures, co-ordinated oversight, formative appraisals, and integrated inclusion of students, internship facilitators, faculty overseers and affiliated industry supervisors through a unified site [27], [30], [31].

These findings point to a gap in research of conducting an assessment of a localized internship management system, aligned specifically according to the institutional situational factors and operational needs of the University of Cabuyao. In this scenario, the paper will be discussing the application of INTERNTRACK as a web-based and mobile-responsive internship management software that is ready to centralize internship database, standardize operations, and assist in coordinating stakeholder activity in internship management.

Based on this, the paper analyzes INTERNTRACK based on how it can be accepted as a centralized digital platform of the administration, monitoring, and supervision of internship at the University of Cabuyao. The system also features student records, attendance validation, submission of journal and documents, progress tracking, stakeholder communication and performance assessment functions. The study uses selected ISO/IEC 25010 software quality characteristics to identify whether these system functions are acceptable in the internship environment of the university or not.

Objectives of the Study

General Objective

The general objective of this study is to determine the acceptability of INTERNTRACK, a web-based and mobile-responsive internship management system for the University of Cabuyao, after its design and development as a centralized digital platform for the efficient administration, monitoring, and supervision of internship-related activities using selected ISO/IEC 25010 software quality characteristics.

Specific Objectives

Specifically, this study aims to:

1. Design and develop a user-centered, web-based and mobile responsive internship management system that:
    - 1.1. Manages student profiles and records through centralized data storage and profile management.
    - 1.2. Supports internship lifecycle management, including placement, deployment, status tracking, and progress monitoring.
    - 1.3. Facilitates internship documentation, including:
        - digital journal submission and review
        - document compliance, submission, verification, and routing
    - 1.4. Provides monitoring and supervision tools, such as dashboards, real-time status updates, notifications, and performance tracking.
    - 1.5. Enables communication and coordination among student interns, PALD Director, internship coordinators, faculty supervisors, industry supervisors, and authorized academic personnel.
    - 1.6. Generates reports and outputs, including performance evaluations, completion certificates, and internship-related reports.

2. Evaluate the acceptability of the proposed system in the light of chosen ISO/IEC 25010 software quality characteristics through the judgement of chosen end users constituted by student interns, PALD Director, internship coordinators, faculty supervisors, partner industry supervisors and authorized academic personnel in respect to:
   2.1. functional suitability;
   2.2. usability;
   2.3. performance efficiency; and
   2.4. reliability.
3. Evaluate the acceptability of the proposed system based on a number of ISO/IEC 25010 software quality characteristics selected by IT experts with software quality assessment background on the basis of:
   3.1. functional suitability;
   3.2. reliability;
   3.3. performance efficiency; and
   3.4. security.

Scope and Limitation

Scope of the Study

This study deals with the conceptualization and creation of INTERNTRACK, which is an online internship management system for the University of Cabuyao designed to improve internships through digitalization and real-time monitoring as well as management based on data. The scope of work is divided into six key processes.

Student Records and Profiling: It offers a central point where all data related to student internships can be stored. It allows managers to categorize the student internship into active, completed, suspended, deferred, or expelled status, noting that the latter must be justified by providing a reason. This is all made possible through a role-based access control mechanism.

Journal Submission and Review: Students have the ability to upload their daily journals, which include details on tasks performed and hours worked along with the Daily Time Records (DTR). The system provides real-time monitoring of industry immersal hours by keeping track of hours gained against the required hours. The HTE supervisors will have to approve the journal entries and DTRs for submission to the faculties. Discrepancies can be found through the system regarding the documents uploaded against manual input of hours.

Placement and Deployment Monitoring: The system handles the process of internship management through the monitoring of MOA and accreditation of HTE. If the MOA expires or the HTE becomes inactive, no deployment is allowed. The system also helps in placement assignment, scheduling of placements, updating of status, and logging of past changes in placements.

Document Compliance and Routing: In this process, everything is digitized from the start, including the narrative report, clearance forms, waivers, and evaluation forms. It replaces the old traditional process where everything was done using paperwork, into a system whereby the paperwork is all handled on a computer system. Once the papers have been verified, students get instant feedback regarding the validity of their papers. It can also be used for bulk uploading or downloading of the documents.

Performance Evaluation and Reporting: Evaluations are created by the system in both formative and summative types through the use of online forms for the assessment process to derive aggregate scores. In case all the hours and the criteria are met, the system issues completion certificates automatically. Analytics and data visualization provide statistics of deployment patterns (most used partners versus those least used), On-the-Job Training (OJT) status tracking using past records, retention rate statistics, and analysis of company performance (engagement rate and student satisfaction among others). All completed internships are stored without hindering system efficiency.

Progress Monitoring and Supervision: Role-based dashboards will be provided by the system for students, coordinators, faculty supervisors, industry supervisors, and the PALD Director that show real-time updates concerning submission status, completion of hours, and adherence. The system monitors the progress of any student that is considered risky due to delay in the hours completion or requirement issues. Scheduling tools are provided within the system for the organization of class meetings related to internship activities like orientation, consultation, and evaluation. Communication capabilities are also integrated into the system to facilitate cooperation between the student intern, PALD Director, internship coordinators, faculty supervisors, industry supervisors, and other academic staff members. Lastly, MISD will be incorporated into the system for integration purposes.

System Integration (MISD Coordination): The study considers integration with the university's Management Information Systems Division (MISD) to enable data synchronization and interoperability with existing institutional systems, enhancing overall efficiency and consistency of records.

Limitations of the Study

The current research takes place in the context of the University of Cabuyao, as well as in the context of the internship setting, process, and users' roles in regards to the suggested INTERNTRACK platform. It will not consider payroll and accountancy operations, human resources management activities in general, matching service jobs unrelated to the internship placement procedure, and involvement with any external agency system that falls outside the domain of internship operations within the agency itself. The research will only deal with development, design, and assessment of the system in the stipulated timeframe, but will not discuss its organizational effects following the implementation in the long run. Moreover, some images and ready-made drawings for the final product are not included in this draft version of the paper, but they will eventually appear in the final one.

There are some practical and contextual constraints of the study, which can influence the process of collecting data, testing the system, and appraising it. The first constraint has to do with the variation in access to the Internet and devices as the INTERNTRACK system is based on them. The second limitation is associated with the availability and sufficiency of official records of institutions to be used in estimating the population size and determining the samples for further analyses. Given the necessity to access institutional databases in the university to collect relevant respondent data, any issues related to accessing the aforementioned data can affect the eventual number of respondents.

Furthermore, the respondents' responsiveness to the needs analysis, system guidance, and formal assessment may take some time, influencing the overall data collection process. However, such practical limitations are an inherent part of any applied research dealing with organizational settings, and therefore they should be viewed as conditions of operation, rather than as methodological imperfections. Finally, this study is constrained to Academic Year 20252026 and not constrained to assessing long-term effects of the system in terms of organization, behavior, and operation of the university.

Conceptual Paradigm

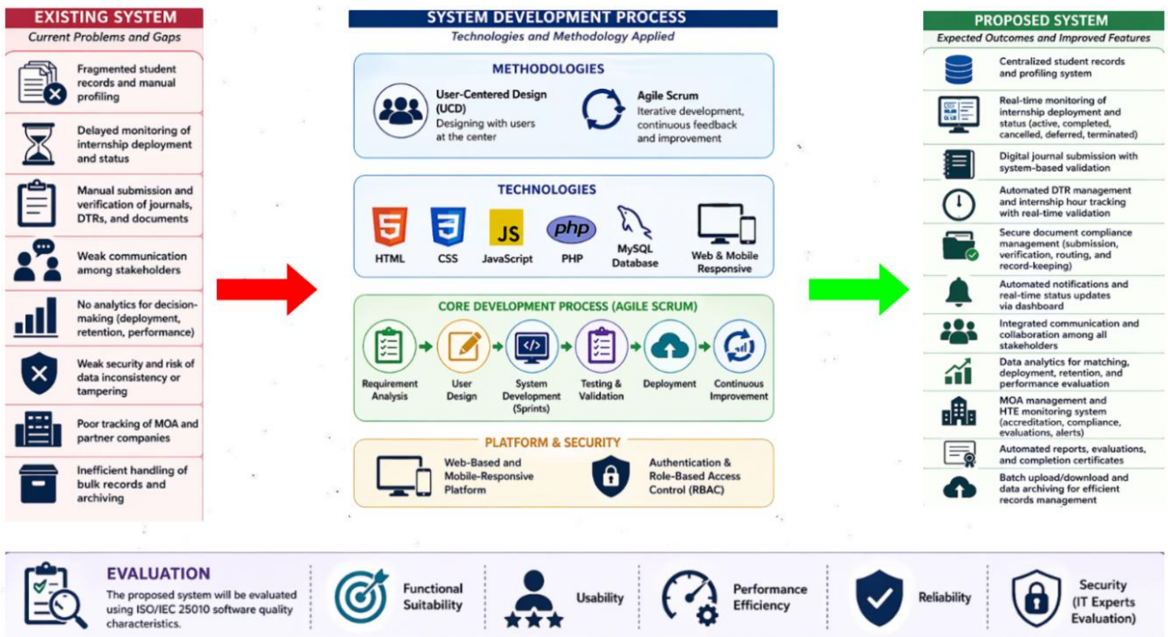

Figure 1: Conceptual Paradigm on the Develop System

The conceptual paradigm of this study outlines how the proposed system aims to transform the current internship management process at the university into a more efficient digital format. On the left side of the framework, we identify the existing challenges faced by the institution, which include disorganized records, inadequate monitoring, inefficiencies in document handling, limited visibility into student progress, and poor communication among stakeholders. These issues highlight a fundamentally flawed internship administration system that relies heavily on manual and disconnected methods. Central to the framework are the technological and developmental strategies designed to address these challenges. The proposed system is guided by User-Centered Design and Scrum methodologies, ensuring that user requirements and feedback shape the system's development in an agile and responsive manner.

The research also incorporates a web-based and mobile-responsive application stack, utilizing HTML, CSS, JavaScript, PHP, and MySQL for data storage, interface development, application logic, and data processing. These technologies were chosen because the issues at hand involve multiple stakeholders, interconnected workflows, and repetitive supervisory and documentation tasks that require close coordination with user activities, all of which will be refined through iterative enhancements of the system's functionality. On the right side of the framework, we present the expected evaluative outcomes of the research, particularly the acceptance of INTERNTRACK based on criteria such as functionality, usability, reliability, performance efficiency, and security. This evaluative framework allows the University to determine whether the system's capabilities—such as centralized record management, placement monitoring, digital submissions, progress tracking, supervisory support, and performance evaluation—are suitable for the institutional context of the University of Cabuyao.

Significance of the Study

This study holds significance because it highlights how internship management can lead to measurable institutional expenses. The current practice of storing records across various documents, spreadsheets, chat threads, and separate files causes faculty and coordinators to waste valuable time comparing submissions and retrieving documents to identify unmet requirements. This inefficiency delays graduation clearances, increases the risk of non-compliance, and hampers the university's ability to effectively monitor students when it is most critical. In response to this pressing administrative challenge in higher education, INTERNTRACK offers a systematic solution that addresses the concrete needs of the University of Cabuyao.

Direction Information Technology in Education. This study contributes to the field of Information Technology in Education by providing quantitative descriptive evidence regarding the feasibility of a localized internship management system, evaluated against the established ISO/IEC 25010 standards. While existing literature often treats digital monitoring and records management as distinct functions, there is a notable lack of studies that present institution-specific evaluation results. These results would illustrate how users assess an integrated platform that combines profiling, compliance tracking, journal reviews, supervisor feedback, and evaluations within a single higher education context. In this regard, INTERNTRACK offers empirical data on the assessment of a locally developed platform, focusing on software quality and its utility at the institutional level. Consequently, this research enhances the current literature on the perceived acceptability of academic technologies that facilitate function-based coordination, comprehensive documentation, and real-time oversight in higher education.

Higher Education Institutions. This study presents evaluative insights relevant to higher education regarding the feasibility of a workflow-based internship management system. It assesses the potential of an integrated platform, such as INTERNTRACK, in areas like requirement tracking, journal submissions, supervisory coordination, and progress reporting. Rather than advocating for the system as a one-size-fits-all solution, the research offers institution-specific evidence on the effectiveness of these functions within a localized internship context. Its significance lies in establishing a foundation for evaluating the applicability of digital internship management in tertiary education.

University of Cabuyao. The research conducted at the University of Cabuyao assesses the effectiveness of INTERNTRACK as a tool for centralizing internship data, managing documentation, and enhancing workflow visibility. The focus of the analysis is to determine if the system adequately consolidates records, monitors compliance, and ensures easy access to internship information within the university's internship management framework. Ultimately, this study aims to deliver tailored insights regarding the system's capacity to aid in the academic administration and organization of digital processes.

Internship Coordinators, Faculty Supervisors and Administrative Personnel. This study aims to deliver quantitative insights to stakeholders about the acceptability of INTERNTRACK as a platform for tracking submissions, conducting compliance reviews, monitoring deployments, and documenting supervision activities. The analysis evaluates how well the system is received in the context of internship administration, focusing on its capabilities to access student records, monitor journal statuses, track attendance, and manage pending requirements. This assessment is crucial as it helps gauge how effectively these role-specific functions align with the oversight

Student Interns. The research aims to provide student interns with evaluative insights regarding the INTERNTRACK platform, specifically its effectiveness for submitting documents and journal entries, monitoring compliance, and receiving feedback from supervisors. This analysis focuses on assessing how well the system's features such as tracking submission statuses, reviewing entries, and accessing returned requirements are received from the end-user perspective. Ultimately, the goal is to determine how suitable the system is as a tool for managing internships

Partner Industry Supervisors. The study provides insights for the supervisors on the supervisory panel within the partner industry regarding the acceptability of INTERNTRACK as a tool for verifying attendance, performance, and communication related to workplace observations. It examines how these functions are perceived in the context of collaboration between universities and industries during the internship period. The significance of this analysis lies in its portrayal of the system as a structured framework for supervisory feedback and outcomes.

Future Developers of the system. The study provides future researchers and system developers with localized quantitative insights into the acceptance of INTERNTRACK within a real higher education setting. While many articles discuss internship systems or present general findings, there is a notable lack of studies that offer evaluations tailored to specific institutions.

Definition of Terms

Conceptual Terms

Automated Document Workflow. This, in this work, means computerized receipt, forwarding, tracking and endorsement of internship papers in INTERNTRACK which has substituted the scattered paper collections and manual strategy of having to arrange [14], [24], [25].

Digital Journal. This is the electronic record in this study where student interns records the daily activities, attendance, time spent, and other internship task related activities to monitor and ensure compliance in this case [17], [22].

Formative Feedback. In this work, this can be taken to mean the continuous comments, guidance and performance related input that is given during the internship period with a view to improving the student interns before the actual assessment at the end of the internship process [27], [30].

Internship Lifecycle. In this work, this means the cycle of the internship activities linked to the student registration and placement up to the monitoring and documentation as well as supervision and final assessment [1], [2], [17].

Summative Evaluation. Here it is used to refer to the ultimate evaluation of the overall performance, accomplishments and adherence of a student intern after fulfillment of the requisite internship duration or hours [1], [4], [27].

Technical Terms

INTERNTRACK. This, in this case, is the intended web-based and mobile-responsive internship management system that has been designed to be centralized to organize internship records, monitoring, supervision and evaluation of the University of Cabuyao.

ISO/IEC 25010. In this paper, this means the international software quality model to determine the acceptability of INTERNTRACK based on the functional suitability, usability, reliability, performance efficiency and security.

Mobile-Responsive System. In this context, this means a system that has an interface that is automatically adjusted to fit varying screen sizes and device types to ensure it can be used on both desktop and mobile platforms.

Role-Based Access Control (RBAC). This can be understood, in this case, as a security system where system functions and data access are restricted based on the role of the authorized user, like student intern, student coordinator, student faculty supervisor, or student administrator.

Scrum. This, in this study, is the agile software development framework that is applied to manage the organization and coordination of development into iterative cycles and integrating stakeholder feedback regularly as the system is created.

User-Centered Design (UCD). In this work, this means the design methodology that consider the needs, tasks and context of users by using an iterative process of prototyping, testing and refining systems to enhance the usability of the system.

Web-Based System. This is used in this study because it means a system that is available on a web browser without the need of having local software installed, which means that it can be accessed by multiple users at different locations and devices.

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents the literature and related studies that informed the development of INTERNTRACK. It is organized into conceptual literature, research literature, and synthesis. Together, these sections establish the theoretical foundation, design considerations, implementation trends, and empirical support for the proposed localized internship management system for the University of Cabuyao.

Conceptual Literature

The conceptual literature focused on theory and design lays the groundwork for comprehending the creation and assessment of internship management systems. Some research [6] through [9], [15] and [16] treats internship administration as a set of discrete clerical tasks, and others view it as an integrated, technology-facilitated process. This includes record keeping, communication, supervision, monitoring and evaluation of the internship process during the whole process. Together the findings show that digital systems do more than simply record, they are also active record-managers and are able to manage the interactions between users, documents, decisions, and timelines that are relevant to the implementation of internships. Moreover, the studies show that there is a conceptual gap between the studies as many have outlined the features needed in an internship management system, but few have given a guide on how to integrate these features into a system that is particular to a specific institution without losing the work flow clarity, accountability and usability. Thus, this paper investigates four closely related aspects that are the conceptual backbone of the design logic of the INTERNTRACK system: the properties of an internship management system, guidelines for development of such a system, requirements for a user-centered system, and testing procedures.

Standard Features of an Internship Management System

Conceptual literature continually defines internship management systems as integrated digital systems that are to enable communication, monitoring, documentation, and evaluation of the internship process. These sources show internship administration as being mostly about information systems, and that timeliness, traceability, among stakeholders and access to records are important. The perspective shifts the understanding of internship management from mere clerical work to one that depends on constant flow of information through different stages of administration. The literature, therefore, suggests that an effective internship platform should, besides conversion, monitor the data management and the workflow.

Common elements identified in the literature include student profiles, monitoring of placements or deployments, documentation attendance, journal submission, documentation compliance, student progress monitoring, feedback systems, and performance assessment [7], [17]–[22]. These elements are common as they reflect the major phases of the internship cycle: student preparation, placement, supervision and evaluation. Upon closer inspection, however, it is found that these elements have different functions. Attendance and documentary compliance are more about control, for example, while progress tracking, feedback and performance evaluation are about supervision, to give but a few examples. The differentiation implies that it is necessary to create an effective internship system that creates institutional accountability, instead of diminishing the internship process.

In the literature, role-based access is known to be a fundamental structural requirement and not just an extra security measure. Evidence shows that clear permissions can increase accountability, simplify work processes and improve the safe handling of academic records. There is also a considerable design constraint in the literature: having a system that allows wide access can make it easy to adapt the system, but may cloud the issue of accountability and introduce the disruption of the order of approvals. In response to this concern, INTERNTRACK provides specific privileges to different roles such as: student interns, internship coordinators, faculty supervisors, industry supervisors and authorised academic staff. It also represents these theoretical principles in the form of separate but related modules for profiling, placement, journal management, compliance tracking, supervision and evaluation. These translate into tangible design requirements for the role based framework of INTERNTRACK and its functional capabilities.

Guidelines for Developing the System

The conceptual literature not only details the necessary elements of an effective internship management system but also offers some guidance on how to construct an effective management system that is maintainable, accessible, and consistent with institutional practices. All the sources [6] [8] [12] [15] [16] studied so far, recommend the same key development principles as modular architecture, centralized databases, and web-based implementation. Some of the key benefits of modularity include the ability to decompose large system functions into smaller ones. Centralized data management is essential because internship records are less reliable when they're spread out in many different interfaces and storage areas. Further, web-based implementation is acknowledged as a successful method to expand access among users in a variety of settings and administrative contexts.

Although the recommendations are useful in general, literature is not clear on the ranking of these recommendations when they are applied in different institutions which have different resources, workflow and user readiness. From a technical perspective, a modular system might seem an advanced approach, but if the modules don't match the sequence of internship activities performed by the parties involved, then it can fail. Likewise, a centralized data repository can increase consistency, but requires a careful design of the processes to make sure that the data recorded is timely, accurate and appropriate for the different roles. This means the principles of development are not just best practices, but need to be put in context of institutional workflow, the duties of the stakeholders and the reporting requirements.

The literature clearly shows that the current literature emphasizes the need to integrate dashboard monitoring and mobile accessibility in the modern concept of internship. Studies show that for both supervisors and students, there is a need to access records, give updates, and make sure they stick to the rules outside the confines of an office environment. The ability to operate from a cell phone is one of the added features that makes this system very user-friendly. One of the key aspects of dashboard monitoring is its ability to convert data into actionable insights, allowing users to easily monitor submissions, deadlines, attendance, and progress metrics in real-time.

Another concern comes to mind at this point. Being accessible on different devices does not mean that it is well monitored. However, if the information is not communicated in a manner that is easily understood and acted upon by users, the system may not perform as well in supporting decision making. Research reveals that there is a major disconnect between the system's technical ability and the utility for management. INTERNTRACK is designed to do just that, with both a web-based and mobile-friendly interface as well as dashboard views to monitor compliance status, track submissions and evaluate progress. In addition, it has a centralized database and modular functional areas, so that tasks like profiling, placing, submitting journals, routing documents, supervising and reporting are interconnected and manageable. The design decisions are in agreement with the literature that the system should be technically sound, and it should be easy to operate and flexible to the administrative requirements of the University of Cabuyao.

Requirements for a User-Centric Design

The literature on the subject underscores the importance of developing internship management systems with a user-centered design approach that accurately addresses the needs, expectations, and tasks of all stakeholders. Various sources highlight that the quality of design is not solely determined by the visual appeal of the interface but also by how well the system mirrors the actual sequence of user activities in real-world scenarios. This aspect is particularly crucial for internship systems, as they cater to diverse groups of stakeholders, each with unique objectives and responsibilities. For instance, students are responsible for submitting their requirements and tracking their progress, coordinators ensure compliance and manage record flow, while supervisors assess performance and monitor advancement. Therefore, a user-centered design approach necessitates a coherent alignment between interface design, workflow processes, and functions tailored to specific roles.

The literature emphasizes that a user-centered design should prioritize clarity, accessibility, simplicity, and ease of use; however, these attributes gain significance only when viewed in the context of specific tasks [10], [13], and [29]. When considered together, these studies suggest that users are more likely to successfully engage with systems when the interface minimizes unnecessary procedural complexities and aligns functions with actual work needs. An interface that is merely “simple” cannot be deemed effective if it obscures essential steps for review, validation, or monitoring that are integral to institutional workflows. In the realm of internship management, therefore, simplicity should be defined not just by visual minimalism, but by its relevance to tasks and the reduction of friction in the user experience.

These findings have significant implications for INTERNTRACK. Student users need straightforward submission tools, features for checking their status, and easily accessible feedback, while coordinators and supervisors require streamlined access to monitoring, reviewing, validating, and evaluating processes. A consistent theme in the literature is that involving stakeholders early in the design process enhances the system's relevance and alignment with institutional needs. Gathering requirements, creating interface prototypes, and consulting users are essential steps, as they uncover local regulations, approval processes, and reporting expectations that generic systems often miss. INTERNTRACK addresses this by tailoring its interface flows and role-specific modules to meet the needs of the internship stakeholders at the University of Cabuyao. As a result, the principles of user-centered design support INTERNTRACK's focus on localized usability, stakeholder-driven interface development, and interaction design that aligns with specific workflows, rather than adopting a generic platform approach.

Testing Procedure

After the importance of user centered design has been established, literature points towards 'Testing' as an important phase for development of internship management systems. Several research studies have highlighted that testing is not only a means of identifying technical errors but also as a way of evaluating the system to provide an organized workflow and to support actual internship practices. With internship platforms, a failure can be a minor software bug or it may be that it takes a long time to be approved, its features aren't available, it lacks in-depth monitoring, your submission is misplaced, or there is poor communication between platform users. This all-encompassing view on testing is especially relevant to systems which involve multiple academic and professional players.

Based on the literature, the criteria which should be considered in the evaluation of internship management systems are criteria that are specific to academic platforms with various roles for users. For it is necessary to measure usability by seeing whether or not students, coordinators and supervisors can easily and clearly do their tasks properly. Further, the usability and responsiveness of the system needs to be assessed, since users may use the system on different devices and at different times and locations. In addition, it is important to test workflows for accuracy; processes like submission of a proposal, review, validation, notification, and evaluation should move seamlessly between stakeholders without missing important steps.

The current body of work does not contain a thorough study of the validation of cross role handoffs in workflow-based academic environments. There is a wealth of discussion on software quality and user satisfaction, but little guidance on assessing approval chains, notification dependencies, and status transitions in environments where various user roles have different institutional responsibility. This is a very big gap because the seamlessness of technical performance and administration is critical for successful internship systems. INTERNTRACK aims at solving this problem by introducing as a measurable criterion the concept of workflow support. The goal of its evaluation is to see if the modules for submitting journals, compliance review, progress tracking, notifications and performance assessment work well in a variety of roles. Therefore, the testing literature supports the evaluation of the service and acts as a technical verification as well as a quality assurance system in the management of internships, offered by the service INTERNTRACK.

The literature in the field has stressed that internship management systems should not be a one-off but should be developed as an integrated, role-based, and workflow-driven system, not just as a repository for students' files. It lays out four important requirements. Firstly, the system shall include the basic functional aspects that involve the entire internship process. Secondly, it is important that its design and implementation be such that it is maintained, visible and accessible. Thirdly, it should be made to match the current work and requirements of the users. Lastly, the system needs to be assessed not only for its technical accuracy but also for its effectiveness in facilitating academic processes involving multiple stakeholders. These common requirements constitute the basic idea behind the development of INTERNTRACK, the web-based and localized Internship Management System for the University of Cabuyao. In addition, they introduced the context for the next section of the research literature which examines application, evaluation and critique of these conceptual standards in previous research.

Research Literature

Review of Related Literature

The reviewed literature converges on the view that internship administration must sustain the entire internship lifecycle through integrated record keeping, monitoring, communication, and assessment. Across the literature, web-based and digital systems are presented not merely as convenient tools but as institutional mechanisms that improve accessibility, coordination, and process visibility among stakeholders. Collectively, these findings establish that effective internship management depends on the alignment of administrative control, supervisory oversight, and timely information exchange. This body of research therefore provides a strong empirical backdrop for the development of INTERNTRACK.

Internship Management in Higher Education

Internships have become an important means of learning, increasingly recognized in higher education as essential opportunities to connect theory to practice in the workplace. Internships have been found to be important in developing workplace skills, preparing for the workforce and linking classroom learning to real world organizational experiences. Therefore, the management of internships goes beyond a side activity of academics and is an integral part of experiential learning that requires careful institutional planning, monitoring and support.

Moreover, the literature has highlighted that IEs require to be managed by HEIs in a holistic and effective manner, and this must involve a range of interrelated components including placement, supervision, documentation, monitoring and performance assessment. These are not administrative tasks that are done in isolation, but are part of a unified academic process. If accountability is not working well in one area, it can affect accountability in another, evaluations, and growth of students. In general, the research presented above shows that effective internship management involves both administrative and educational aspects, with a focus on ensuring that the experiences of interns are always aligned to their academic standards.

The effectiveness of internships is found to depend not only on student participation, but also the overall internship process is managed by a consistent and organized manner by the institutions. Research has indicated that optimal internship administration correlates with timely completion of internship requirements, effective management of paperwork, and communication among all stakeholders. In contrast, internships can be of exceptional quality, but not benefit students' education if the placements are poorly coordinated. The evidence makes it clear that establishing an “internship management structure” where guidance and stability are provided simultaneously is especially crucial in the rationale for INTERNTRACK.

Digital Management of Internship Records and Documents

Traditional manual record keeping is being replaced by digital management of internship related information, which is highlighted by the existing literature. Overall, the research studies in [14, 17, 24 and 25] show that digital systems are very helpful in the management of submissions, reduce delays caused by disorganized documentation and further enhance the organization of the internship records. Rather than just a document repository, these studies portray digital record management as a management tool where one can access information faster, see statuses more clearly, and ensure reliable monitoring of internships.

This transition is especially significant because in the internship administration, effective document management plays a very important role. It is important to note that students are required to submit different forms, journals, clearances, and evaluation records at various points during the internship process [14], [24] and [25]. With the help of digital systems, coordinators and supervisors can check, monitor and evaluate submitted documents more consistently. This means that problems with follow-ups, missing documentation, or incomplete compliance are eliminated, which increases transparency in administrative documentary processes and for student interns.

These studies suggest that the control of digital records and documents can improve the efficiency of the institutions by increasing accessibility of the records and documents, making them consistent and ensuring better coordination among users. Rather than using paper forms, unorganized spreadsheets, unmanaged files and documents that need to be manually updated, a single digital system allows all stakeholders to work from a single source of information, so they can respond faster to any missing or delayed requirements. This is a joint research that serves as the basis for determining the effectiveness of using the modules created by the project, namely the module of document compliance, module document submission and module of centralized internship records, as part of the university documentary control system.

Monitoring, Supervision, and Communication in Internship Programs

From the literature, it is seen that the management of internship is dependent on continuous monitoring and supervision during the internship period. Monitoring is described as a continuous process in studies [17], [22] and [27] which includes attendance, activity and review of the work journals and observation of workplace performance. This is important because it refutes the idea that the oversight of internships can be delegated to a one-time validation or to an end-of-term assessment. The literature, however, places monitoring in a different perspective, as a process that institutions can use to keep them accountable and ensure that the internship experience is of quality educational value.

The literature together also indicates that digital technologies contribute to enhanced supervision, thus, allowing coordinators and supervisors to monitor progress in a timely and systematic way [22], [27] and [30]. The digital validation features, progress dashboards, and online records provide a consistent picture of student activity for faculty and industry supervisors and enable a faster response to any problems. This allows the supervisions to be more evidence-based, as the decisions and follow-up actions can be based on available records and not be left with reports or lack of updates.

In the literature, communication is also recognized as a key factor for successful internship administration and it entails interactions between students, coordinators, faculty supervisors, administrators and partner industry supervisors [17], [27], [30], [31]. The literature comes together to say that poor communication channels contribute to delayed reporting, validation and feedback, which ultimately leads to poor internship management. In contrast, systems with communication integration to monitoring increase collaboration and minimize process disruptions. The following pattern is documented in literature and is the basis for an evaluation of whether or not the integrated functions of progress monitoring, supervisory review and communication with stakeholders are deemed acceptable in a single platform.

Web-Based and Mobile-Responsive Academic Systems

The literature clearly indicates that the implementation and supervision of an internship depends on continued monitoring and supervision throughout the internship period. Research studies [17] [22] and [27] depict monitoring as an ongoing duty that involves checking on attendance, reviewing activities, reviewing journals, and observing performance in the workplace. This view is very important because it rejects the notion that supervision can only happen once or at the conclusion of the internship. Instead, literature portrays monitoring as an ongoing activity that will help institutions maintain accountability and keep the internship experience educationally intact.

Furthermore, the literature shows that digital technologies improve the supervision process, enabling the coordinator/supervisor to monitor the student's progress on time and in an organized way [22], [27] and [30]. Digital validation systems, progress dashboards and online records allow faculty and industry supervisors to review and deal with any issues more quickly and at a more consistent rate. As a result, supervision becomes more evidence based with decisions being made and follow-up actions taken, based on readily available records rather than delayed reports and incomplete data.

All of the above research indicates that the adoption of an online and mobile-friendly platform allows users to file and deliver materials on time, respond to information, and download it faster and more efficiently. The mobility associated with the elimination of place and/or device limitations makes internship management more flexible and continuous. This congruence in the literature provides a starting point to evaluate the potential of a web-based and mobile responsive system (INTERNTRACK) for accessing internship records and managing tasks and submission processes in real-life settings where stakeholders operate.

Usability, System Quality, and User Experience

There is a strong emphasis in research that usability and user experience are not only ancillary design features, but key components that shape the success or failure of a system. Research on system design [5], [10], [13] and [29] has shown that users are more likely to accept a system that is intuitive, easy to use, and oriented towards their tasks and requirements for information. Altogether, these results confirm usability as a key feature that can impact ease of use, adoption of the system, stakeholders' trust and the real-world benefits of the platform in the institutional settings.

Studies [10], [13] and [29] underscore the importance of user-centered design in the quality of interfaces and the effectiveness of a system. Incorporating end-users into the requirements, prototype, and design process allows developers to design systems that better represent the process of the institution and that satisfy the expectations of the end user. It is especially important for internship management as different coordinators, supervisors and students use this platform, and their needs for access, feedback and engagement are different. Overall these findings suggest that system quality is shaped by both the technical strength of a system, but also by design decisions made to reflect the real-life use patterns of stakeholders.

Furthermore, studies on system quality and user experience suggest that issues like accessibility, reliability, and security are critical to users' evaluations and trust in digital systems. If the system does not meet the practical needs of the user because of its failure to be reliable, easy to use or secure, it may be technically correct, but not very useful. The literature, therefore, highlights the importance of designing internship management systems that are user-friendly, understandable, simple, reliable, and trustworthy. This line of research acts as a starting point to evaluate the usability and software quality requirements for its target users within a multi-stakeholder academic environment that is expected of the system, namely, INTERNTRACK.

Review of Related Studies

The reviewed studies collectively validate the effectiveness of internship-related systems that integrate digital s, document management, attendance tracking, progress monitoring, and evaluation tools. Rather than treating these functions as isolated features, the empirical literature shows that their value emerges when they operate within a coordinated platform that supports supervision, compliance, and record visibility across the internship lifecycle. The findings therefore shift the discussion from simple digitization to institutional process control, demonstrating that well-designed internship systems strengthen both administrative oversight and stakeholder accountability.

Studies on Internship Management Systems

These systems have been repeatedly emphasized in internship management system research as integrated solutions that streamline internship processes in educational institutions. The results of different studies show that these systems have the ability to reduce administrative work, which is usually divided into various processes like profiling, registration, deployment, documentation, and monitoring tasks, to a single system. The integrated approach enables institutions to realize more consistent process flows, better continuity of records and better supervision of records.

Furthermore, developmental research in this area highlights the importance of building formal systems that are derived from a detailed assessment of the needs of the institutions, changing the form of the system in an incremental fashion, and testing the system repeatedly. These studies are not only about the methodology, they show that systems created based on a requirements-driven approach better serve the roles of users, their workflows, and the institutional goals and aspirations. The trend highlights that internship platforms can't simply be bought off the shelf and must be customized to match the operational needs of their context.

The studies highlighted suggest that it is important to ensure that roles of users are matched with the institutional processes to make a system effective. Systems that take into account local practices, reporting mechanisms, and permissions based on roles are more likely to operate smoothly, with each participant operating according to their assigned role. This is observed trend which supports the evaluation of INTERNTRACK as a customizable system as its modules, permission and workflow are based on the University of Cabuyao internship processes.

Studies on Digital Attendance, Journal and Monitoring Systems

Various related research studies [17], [20], [22], [27] and [30] support the positive outcomes of digital attendance systems, activity logging, and tracking progress during an internship. The findings showed that e-Attendance and e-Attendance Journal systems were highly effective in enhancing the accuracy and timeliness of student activity records. The improvement allows institutions to track compliance more effectively than with traditional methods which depend on reports that are only submitted at a later date or that are only compiled manually. These digital systems don't just address the lack of visibility, which is a common issue during internships; they're designed to fill in the gaps that can make it difficult to respond to issues in time.

Moreover, the result of these studies collectively demonstrates that digital monitoring tools improve their supervisory responses by providing students with the ability to update them regularly and by speeding up the process of reviewing work by faculty or coordinators [17], [22] and [27]. Results reported aren't just better recordkeeping, but also more accurate progress monitoring, better tracking of hours and work completed, and earlier detection of missing or out-of-order work. Accordingly, the literature highlights the importance of monitoring features to enable ongoing supervision, rather than just serve end-of line validation.

Furthermore, studies of online and mobile monitoring systems also confirm that access to student information on demand is crucial to improving accountability and oversight. Real-time (or near real-time) data provides supervisors the opportunity to solve problems quickly, like delays, missing entries, and performance issues, regardless of the users' location. The results are used as a basis for evaluating the effectiveness of the digital journals, attendance monitoring and progress monitoring which are part of the study as supervisory tools.

Studies on Document Workflow and Compliance Systems

Each study on digital document workflows and compliance systems has stressed the need for the processes of submission, validation, routing and approval to be well organized, and to be greatly enhanced by digital compliance and process management. Based on the results, this provides a means of identifying inefficiencies that exist when using repetitive document tasks in these systems and how they help to streamline the submission process, shorten review timelines and improve the visibility of the document status for students and administrators. As a result, compliance management is more traceable and less reliant on fragmented follow-up activities.

Further, the research indicates that document workflow systems will facilitate institutional monitoring by bringing together the documents into a single, readily accessible platform, enabling authorised individuals to be sure that the required documents are submitted, reviewed and verified. The system reduces problems like lost documents, late delivery of documents and irregular retrieval of records, and it also enables better visibility of the state of the submitted requirements. These results hold significance as they directly tackle common administrative issues like missing documents, unclear submission statuses, and delayed validation procedures.

Furthermore, document compliance systems, such as those found in [14], [17], [24] and [25] provided a more unified and comprehensible documentation process for the student and supervisor. Students gained better clarity on incomplete submission and outstanding requirements, and coordinators and supervisors had the opportunity to review student records via a more streamlined compliance framework. The results of these findings are used as a basis to evaluate the effectiveness of the digital document submission, routing and validation functions of the INTERNTRACK system in terms of their procedural transparency and document management.

Studies on System Evaluation and User Acceptability

Numerous studies of system evaluation and user acceptance have shown that the only way to ensure that a system will be effective is to make sure that the user is willing to accept it. The studies highlight the need to consider usability, utility, reliability and user satisfaction by directly involving experts and target users. Results show that all technically functional systems need to be fully assessed to determine their performance in the context of the institutions where they are implemented.

The studies reviewed tended to use usability surveys, expert ratings, and testing with users to determine the effectiveness of the built system to accomplish the intended system goals. These approaches identified the system's interface issues, addressed user confusion, and assessed its usability and real-world use beyond mere functionality. Thus, evaluation is seen not only as a formality, but as the necessary step to establish the quality of the system and its effectiveness in functioning.

Studies have shown that responsiveness, security, accessibility, and interface clarity are some of the factors that affect user acceptance. Systems that fit into users' workflow, perform consistently and build trust in data management and functionality will be more readily adopted. This set of findings, together with others, enables the formal application of evaluation criteria in INTERNTRACK, specifically in determining the technical and usability criteria the system must adhere to for the target users.

Implications of Related Studies to the Present Study

All the studies reviewed showed a positive effect when internship systems were embedded in a unified operational framework with responsive access, role-specific permissions, centralized data management, and monitoring. These findings not only confirm that digital is better than manual, but also identify the exact mechanisms that make digital more effective, including better role clarity, faster validation processes, greater visibility of progress and better coordination of stakeholders. This empirical evidence thus strongly supports the development of a platform that brings together record keeping, supervision, compliance and reporting into one system architecture.

Furthermore, the literature points out that some issues like institutional compatibility, user preparedness, and protocol compatibility are essential for the success of these systems. If there are differences in roles, documentation, communication, and approval, a system that has been validated in one context may not easily transfer to a different context. This is of special significance in the present study since there is little literature that highlights locally specific enlightenment into internship management systems which correspond to the specific handling procedures laid down in the University of Cabuyao in the Philippines. The study therefore sought to examine the acceptability of INTERNTRACK as a localized system of internship management within the specific institutional setting as a descriptive-evaluative study.

Overall, the research reviewed indicates that mobile access, constant user engagement, localization and a holistic monitoring and oversight strategy are essential to the relevance of systems [18, 19, 27, 30, 31]. While there has been research on some of these areas, such as automation, compliance monitoring, and digital access, the integration of coordinated supervision, constructive feedback and customized process design for individual institutions has not yet been researched. This identified gap directly influences the development of the INTERNTRACK, the purpose of which is to integrate the processes of monitoring, communication, compliance management and role-specific supervision that is suitable to the unique operations of University of Cabuyao.

Synthesis

The literature and studies reviewed uniformly indicate that the internship process is not just a mechanical clerical task but is a complex academic activity that requires careful management of internship records, submissions, internship supervision, communication, and assessment. Theoretical frameworks emphasize the critical elements needed to be in place for such systems, including a focus on user-centered design, integration, aligning workflows, and systematic testing. Empirical studies have also verified that implementing these principles has tangible benefits for institutions when applied via role-based access, digital platforms, document workflow systems, tools to monitor progress and structured evaluation processes. Theories and empirical findings show that good internship systems are ones that synchronize technical skills with the actual institutional processes of development.

This literature focuses on the digitisation of internship processes, rather than on how specific functions of the system solve specific administration and supervision problems. For example, role-based access improves accountability and digital attendance data makes it easier to see student engagement. Document workflow systems can minimize delays in document submission and compliance concerns, and mobile-friendly access extends the reach of supervision and validation. But there is also an important lesson from the literature, that is, these systems only work when they are localized. They function best when they support the real roles, approvals, documentation requirements and communication processes of the institution they are put into place in.

Thus, INTERNTRACK is an important topic for theoretical and practical study. The literature provides a description of important design principles that a credible internship platform should observe, and research suggests that practice of these principles is associated with concrete improvement in institutional practices. One of the shortcomings of this body of work, however, is the lack of detailed documentation in the local context of the specific protocols, supervisory structures and academic culture of the University of Cabuyao. This paper is designed to bridge this gap, by assessing the acceptability of a locally developed, web-based and mobile friendly internship management system based on selected Software Quality Attributes (SQA) from the ISO/IEC 25010 software quality standard.

CHAPTER III
METHODS AND PROCEDURES

Introduction

The methodologies and procedures used in the study are explained in this chapter, which covers the research design, the participant's characteristics, the research location, the sampling of the population, the data collection process, statistical analysis, the preparation of the system and the ethical considerations. They create the basis for the quantitative and descriptive evaluation of INTERNTRACK.

The methodologies discussed here are designed to fit the purpose of the study which is to evaluate the acceptability of the INTERNTRACK at the University of Cabuyao, using the right population for the study and the appropriate ISO/IEC 25010 criteria. The chapter focuses on the use of structured questionnaires and descriptive statistics to collect ratings from end-users and IT experts, as well as introduce Scrum and User-Centered Design as preparatory frameworks that were used before the main research evaluation.

Research Design

The method of this research is quantitative with descriptive type. It is considered qualitative because data on INTERNTRACK is obtained by using structured questionnaires and presented in a numerical form using descriptive statistics. The study is descriptive in nature because it is looking at the acceptability level of INTERNTRACK as perceived by its end users and IT professionals, using the specific set of software quality attributes of ISO/IEC 25010.

This methodological decision is appropriate as the research is not based on manipulation of variables or causal relationships. Instead it focuses on how the participants rate the system with respect to functional adequacy, usability, reliability, performance efficiency, and security. Thus, this design could be used as a template for evaluating the acceptability of INTERNTRACK in the University of Cabuyao.

Respondents of the Study

This study will involve the key stakeholders involved in the Internship processes in the University of Cabuyao that will be enhanced by INTERNTRACK. This group includes the PALD Director, student interns, internship coordinators, faculty supervisors, supervisors from partner industries, IT staff designated by an academic supervisor, and designated IT staff. These people have combined to present themselves as both the users and technical evaluators capable of determining the effectiveness of the proposed system.

The respondents will be split into two separate groups of evaluators. The first group are end users including the PALD Director, student interns, internship coordinators, faculty supervisors, partner industry supervisors and authorised academic personnel. This group will evaluate INTERNTRACK for its functionality, usability and performance efficiency. The second group consists of IT experts with a background in software quality assessment, who will assess the functional suitability, reliability, performance efficiency and security of the system.

The respondents groups selected are appropriate as they are the intended user groups and technical evaluators of the INTERNTRACK in the university's internship program. They are needed in their involvement to gather qualitative information through questionnaires and to evaluate the system acceptability in term of software quality attributes that have been defined in ISO/IEC 25010.

Research Locale

The University of Cabuyao is located in Katapatan Mutual Homes in Barangay Banay-Banay, City of Cabuyao, Laguna. This site will be the location for the internship experience required of its students, with the following recommended approach to address the needs of its interns.

This location is perfect since it contains the organizational surroundings, the age of the users, the normal running procedures, and the communication process that will influence the proposed system. The studies are carried out in a real clinical environment, allowing the researchers to see real processes, identify problems with the way they work and develop an appropriate system to address those.

The suggested system is different from other internship system designed for general use in different industries as it is being designed in implementing the internship program in the different academic program of the Cabuyao's University Colleges and Departments. This focus enables the study to better match the special workflows and needs of the University.

Population of the Study

The study will utilize all the groups of players involved in internship activities in the University of Cabuyao in the academic year 2025 – 2026. This encompasses the PALD Director or representative from an equivalent office, internship program coordinators, faculty supervisors from Internships from various academic programs, industry supervisors, authorised academic staff (e.g. internship advisors) and student interns involved in the internship programs. In addition, the technical quality of program documentation will be used to assess IT experts participating in the internship programs.

The number of participants in each category will be decided by appropriate University offices, colleges and internship coordinators based on the official records of the institution. These will be used to determine the overall size of each group and to determine the sample size if needed.

This population includes the university students doing internships, the staff members who have the responsibility of assigning them and supervising their work, as well as the lecturers who are responsible for judging the outcome of their internships. The proposed system could be implemented in the real internship environments in the university so that this group can test the system in real scenario within the university. In addition, it contains everyone and every unit that is involved in the internship process - such as those who submit the internship requirements, monitor the internship development, review documents, oversee the internship, validate internship results, and evaluate internship outcomes - who use the system on a daily basis.

Sample Size Determination

The number of respondents for each group will be determined by the nature and size of the group. A total enumeration approach will be used for smaller, more manageable groups, such as PALD Directors, or their equivalent, internship coordinators, faculty supervisors, academic personnel authorized to award academic credit for internships (hereafter “academic personnel”), and IT staff, knowledgeable about internship administration and system quality (hereafter “IT staff”). This is because of their specialized purposes and the insights they can provide on how to provide the best management of internships, and how effective the existing system is.

For larger groups (e.g. student interns, and, where possible, a few industry supervisors from partner organisations) the minimum number of members will be determined after receiving official population data from the appropriate university departments or partner companies. If the population is not large enough to be sampled comprehensively, Slovin's formula will be used to determine the sample size for a specific margin of error and then the sample will be proportionally distributed among the strata of interest.

Based on official population numbers, the relevance of each of the respondent groups to the specific phase of the study and the feasibility of the study team to gather the required data from each of the respondent groups within the fieldwork timeline, the final sample size for each phase will be determined.

Sampling Technique

A purposive sampling technique with a non-probability sampling model would be used to select the participants of the study. The selection will be made on the basis of the role and responsibilities, as well as expertise and direct experience of the individuals with the internship system and processes being studied. This approach is suitable as it enables the researchers to obtain pertinent and trustworthy insights from those best equipped to meet the study's aims.

The respondents selected include the PALD director, internship program coordinators, faculty supervisors, IT personnel, student interns and supervisors from partner industries. These are selected by design because they have had first-hand experience and understanding of the internship program and its operations. The use of purposive sampling ensures that data collected will be meaningful, accurate, and relevant to the research objectives.

Sample Distribution Table

Table 1 presents the working framework for the sample distribution of respondents. The final population counts and computed sample sizes will be inserted once the researchers obtain the official records needed for supported computation and final allocation.

Respondent Group	Population	Sample
End Users (PALD Director, student interns, internship coordinators, faculty supervisors, academic personnel)	51	51
IT Experts	5	5
Total	56	56

Table 1: Sample Distribution Table

Data Gathering Procedure

The data collection process will center on a systematic assessment of INTERNTRACK through the use of validated survey tools. Once the required approvals are obtained, the researchers will distribute distinct questionnaires to both the designated end users and IT specialists. This procedure will encompass the preparation and validation of the instruments, orientation for respondents to ensure proper system usage, administration of the questionnaires, collection of completed instruments, and subsequent statistical analysis of the findings.

Type of Data to be Used

The type of data sources to be used in this research are primary data with quantitative approach and secondary data. The main data will be gathered by end user and IT Professional structured questionnaires to assess the acceptability of INTERNTRACK using specific ISO/IEC 25010 software quality attributes. Moreover, secondary data sources will comprise relevant university documents, internship reports, manuals, policy documents and academic literature to help contextualize the study's institutional structure.

The secondary data will be obtained from information from university records, internship forms, manuals, policy documents and relevant literature. The resources will provide crucial documentation for the needs assessment, serve to clarify current practices, and strengthen the theoretical and methodological aspects of the research.

Needs Assessment

Prior to the development of INTERNTRACK a needs assessment will be conducted. The purpose of this stage is to identify the current problems, constraints, inefficiencies and necessary requirements of the current internship management processes of the University of Cabuyao. The results of this assessment will be used to design the features, flow and scope of the functions for INTERNTRACK before it is formally assessed.

Researchers will secure approval from the appropriate university departments before taking data. After obtaining the permission, they will finalize a self-designed structured questionnaire, identify eligible respondents and determine the time frame for data collection. This instrument will be aligned with the study objectives and the identified problem areas in the initial investigation and will address areas like, record accuracy and completeness, timeliness of monitoring, efficiency of document processing, visibility of student progress and communications amongst stakeholders.

The questionnaire will be validated by experts before distribution, to ensure it is clear, relevant and comprehensive. Once validated the instrument will be given to the selected respondents. The responses collected will be collated and analyzed with descriptive statistical analysis (frequency counts, weighted means and rank ordering) to determine the priority needs and requirements to be incorporated into the development of INTERNTRACK.

System Performance Evaluation

After INTERNTRACK is deployment ready, an assessment of the performance will be conducted to measure its acceptability. This assessment will be on two sets of respondents, end user and IT personals. The end users (student interns, the PALD Director, internship coordinators, faculty supervisors, industry supervisors and authorized academic staff) will evaluate the system on the basis of functional suitability, usability and performance efficiency. On the other hand the IT professionals will concentrate on the functionality, reliability, performance efficiency and security of the system.

Selected ISO/IEC 25010 software quality attributes will be used as the basis for the evaluation tool and the indicators will be validated before use to ensure they are clear, relevant and suitable for the system being evaluated. Before respondents complete the structured evaluation questionnaire, they will be given guided access to INTERNTRACK, as well as extensive training in the use of the functionality of the system.

The responses that are gathered will be arranged, tabulated and analyzed with descriptive statistical techniques such as weighted mean to determine the acceptability of each criterion and the acceptability of the whole system. The results will provide quantitative information on the acceptability of the intervention, based on the score given by the different respondent groups.

Statistical Treatment of Data

The study will employ descriptive statistical methods to analyze the quantitative data collected. Frequency and percentage metrics will be utilized, as appropriate, to outline the characteristics or distribution of the respondents. To assess the identified needs from the needs assessment and the acceptability levels from the system performance evaluation, a weighted mean will be calculated.

The weighted mean results will be interpreted according to the five-point descriptive scale presented in Table 2. Additionally, rank ordering may be applied to identify which needs, features, or evaluation criteria received the highest and lowest ratings. Given that this is a quantitative and descriptive study, the analysis will focus solely on numerical summaries and a descriptive interpretation of the findings.

Scale	Range	Interpretation
5	4.21-5.00	Strongly Agree
4	3.41 – 4.20	Agree
3	2.61 – 3.40	Neutral
2	1.81 – 2.60	Disagree
1	1.00 – 1.80	Strongly disagree

Table 2. Five-Point Likert Scale for System Evaluation

System Development Methodology

Introduction

INTERNTRACK is developed using Scrum methodology alongside User-Centered Design (UCD) before the research evaluation takes place. UCD is utilized to ensure that the system's interface, functionalities, and workflow are tailored to meet the needs, roles, and working conditions of users involved in internship administration. Meanwhile, Scrum facilitates the organization of the system's development through iterative cycles, enabling the refinement and completion of features prior to formal evaluation by respondents.

The integration of UCD and Scrum is particularly suitable for INTERNTRACK, as it is designed to be role-based and workflow-oriented within the context of the University of Cabuyao. However, in this study, these methodologies are not regarded as the research design itself; rather, they serve as the preparatory framework for creating the version of INTERNTRACK that will undergo quantitative and descriptive evaluation.

Overview of the Chosen Methodology

A general overview of the methodology that was chosen for the research. In developing INTERNTRACK, User-Centered Design (UCD) has an important role because the interface, process and functionality of the system is designed to suit the needs, activities and context of the user. The research team used UCD to identify key requirements for internships, which were used as the basis to improve the features and interaction process of the system. This helped to develop a user-friendly, role appropriate and goal oriented system.

Further, Scrum is used as a way to organize the development of INTERNTRACK into "sprints" that are short cycles of development. In every sprint, prioritized features are implemented, evaluated, and evolved to make the system more complete and up for formal testing. This approach enables the research team to decompose the preparation process into a series of smaller steps, and to enhance the system step by step.

It is quite appropriate to integrate UCD with Scrum, because the application is a localized, role-specific, workflow-based internship management system, specifically for the University of Cabuyao. Yet, UCD allows for stakeholder needs to be addressed and Scrum provides an organized and flexible way of preparing a system. These methodologies are explained here mainly as a preparation framework for the development of INTERNTRACK, before critically assessing the study quantitatively and descriptively.

Implementation of the Methodology

The following logical steps will be carried out in the selected methodology, converting the needs of the stakeholders into a functioning system. While this text-only version does not include the actual diagrams, the following phases outline the production of the models, interface designs, and software outputs required to develop the models.

Phase 1: User Research and Requirements Analysis

The project will begin with the assessment of user needs and documentation of the existing processes in the University of Cabuyao in the management of internship programs. The researchers will examine current challenges, delays, documentation, and processes, user responsibilities, and any other features that the system might require.

The phase focused on the definition of functional and non-functional requirements will thoroughly describe all the specifications needed for the new system, as well as the involvement of each stakeholder. Moreover, it will generate a priority list of issues the future system needs to address.

Phase 2: Concept Design and System Planning

Once the requirements phase is complete the researchers will go into the concept design and system planning stage. This stage will involve developing the team's thoughts on how to break down the system into key components; role-based access structure; system processes for developing records and submitting records; and initial database needs.

In addition, the researchers will complete the technology stack for the study, which will be largely comprised of HTML, CSS, JavaScript, PHP, and MySQL for a database. The outputs of this phase will include the system architecture, module layout and an initial development plan for features.

Phase 3: Prototyping and System Modeling

After the concept and the corresponding documentation has been developed by our team, the researchers will develop and display wireframes for the information system, different interface concepts and systematic models to guide the development process. They will create use case diagrams, activity diagrams, sequence diagrams, system architecture, data flow diagrams and entity relationship diagrams for the information system as well as wireframes for user interfaces.

This document provides a preliminary overview of the artifacts used in the REQUIREM (ReQUIRE Method) from a methodological point of view. It does not emphasize templates with numerical information but uses these artifacts to focus on how they are used. The purpose of these artifacts is to map out the requirements into several models that demonstrate the users, processes, data flow, interactions and architecture of the system.

Phase 4: Sprint-Based Development

The team will build INTERNTRACK in a series of SCRUM based sprints. In order to help achieve incremental progress, the developers and researchers will identify and prioritize the features of INTERNTRACK into a number of development cycles or sprints. The focus of these sprints will be on key modules, such as student information, user administration, placement management, journal submissions, document compliance and supervision, notifications and program evaluation.

The architecture is crafted to enable gradual development, initially emphasizing key functionalities. There will be a review session at the end of each sprint, where the elements completed will be evaluated and feedback and lessons learnt from any challenges will be applied.

Phase 5: Testing, Refinement, and Evaluation Preparation

Once the basic elements of the system are in place, a process of internal testing and refinement will begin to ensure the system's functionality, reliability, and responsiveness. If any faults, interface glitches or any other process irregularity are found by the researchers, they will be corrected before the system is evaluated by the researchers.

The aim of this phase is to complete a tested and validated assessment tool, testing environment and user manual for performance testing by experts and users. In addition, an improved version of INTERNTRACK will be developed for implementation and testing.

Use Case Diagram

The use case diagrams of the INTERNTRACK system outline the scope of the system, interactions between users and the system, and the roles played by the users of the system. The use case diagrams show how the academic staff manage the requirements of the system, track the progress of the internship, and evaluate the internship, as well as how the coordinators and supervisors approve, monitor, and give feedback on the internship, and how the students search for internships, upload documents and logs, and view the evaluations. These diagrams are also a basis to determine the functions the system is needed for, by showing processes and workflows of the system. Through regular reference, developers and users can help make improvements, optimize performance and ensure a user-centric design.

This diagram describes how a Student Intern can adjust their profile, and view the status they have, and how an Internship Coordinator can review the profile and assign status changes with reasons. Once all updates and reviews are made, the Authorized Academic Personnel will determine if the student's profile is finalized and approve it.

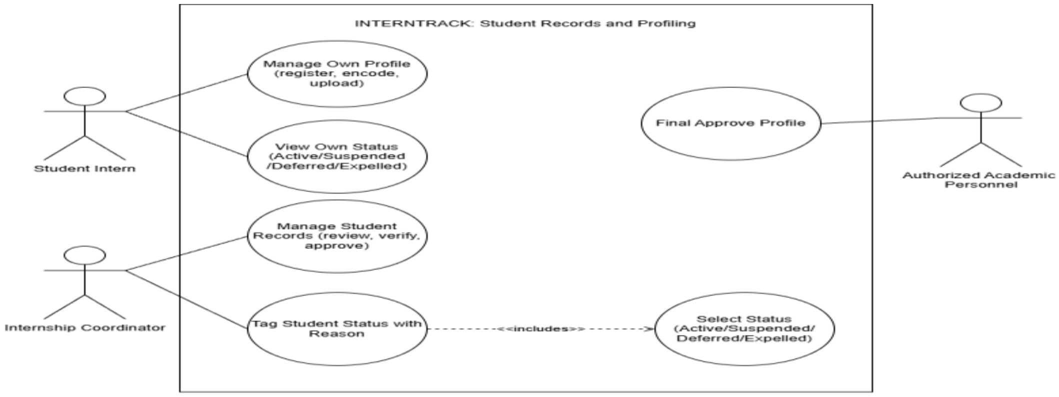

Figure 2. Use Case Diagram for Student Records and Profiling

This diagram illustrates the process of entering journal entries with DTR and hours by a Student Intern, and being able to see the real-time DTR validation of logged hours. Entries are monitored by the Internship Coordinator and validated by the Partner Industry Supervisor to ensure entries are compliant and consistent with the hours recorded.

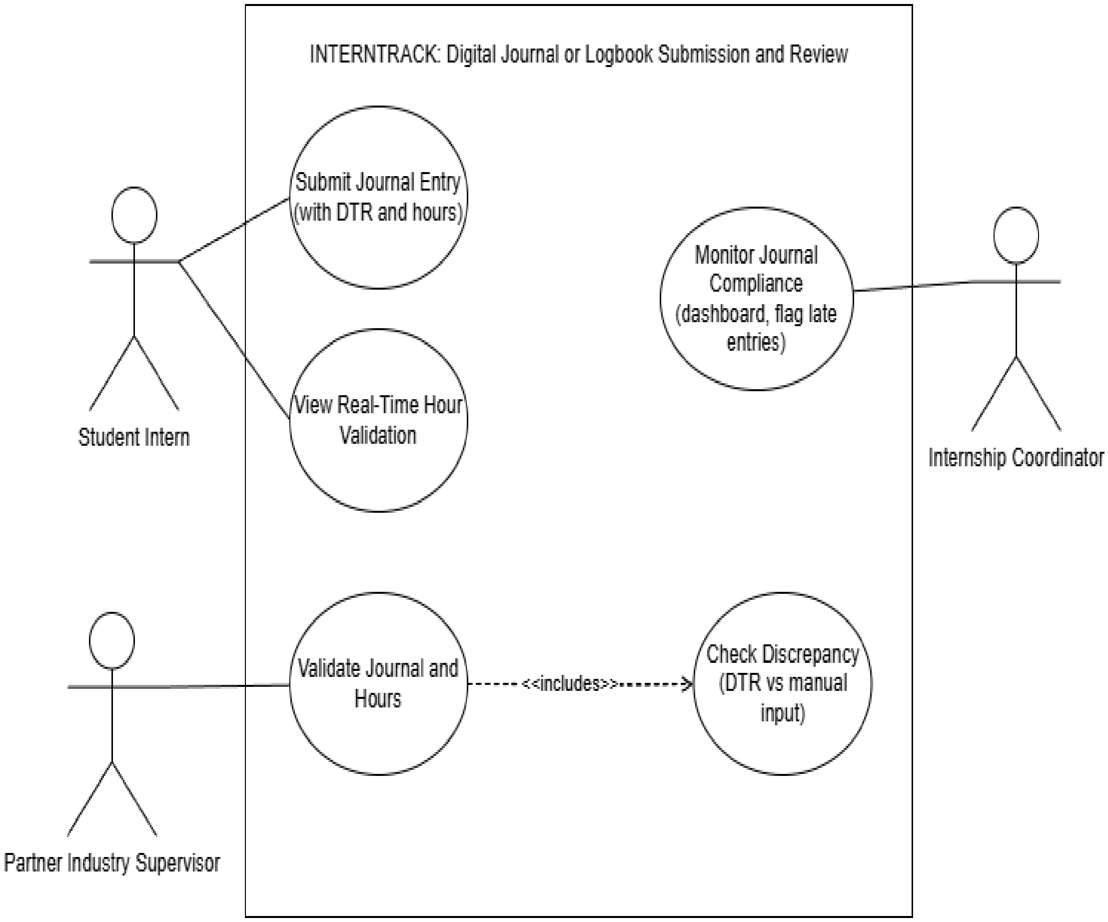

Figure 3. Use Case Diagram for Journal Submission and Review

This diagram illustrates how the Internship Coordinator oversees student placement and deployment, as well as monitoring deployment progress and checking on MOA and HTE status. MOA is not active, deployment may be prevented; Student Intern can check their deployment status and placement.

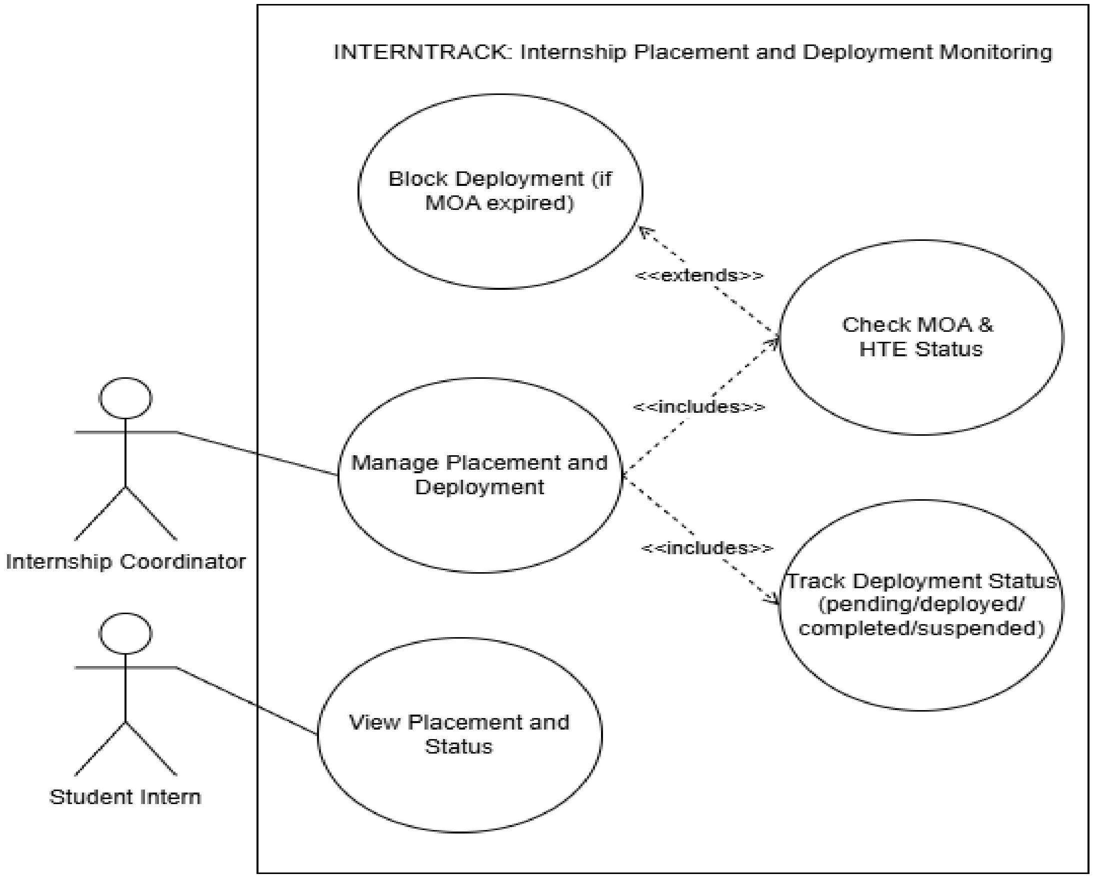

Figure 4. Use Case Diagram Internship Placement and Development Monitoring

This diagram illustrates the process of student placement and deployment by the Internship Coordinator, and the monitoring of MOA and HTE status and deployment progression. The Student Intern is able to see their current placement and status, while deployment is blocked if the MOA has expired.

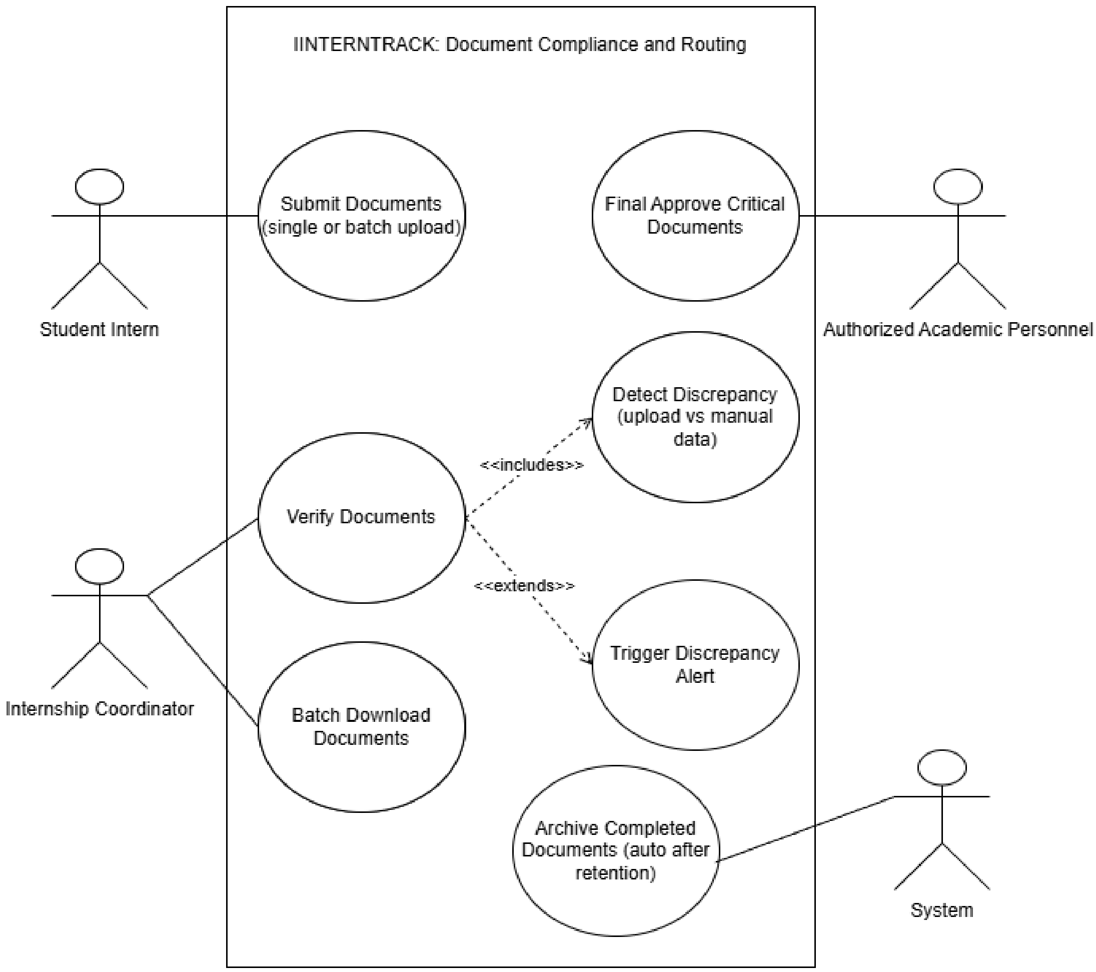

Figure 5. Use Case Diagram for Document Compliance and Routing

Use case diagram for INTERNTRACK system to demonstrate the interaction amongst the various users of performance evaluation and reporting. The Internship Coordinator creates analytics reports, such as the retention rate, deployment trends and company frequency; supervisors rate interns and view results; and the PALD Director can monitor OJT status through a dashboard.

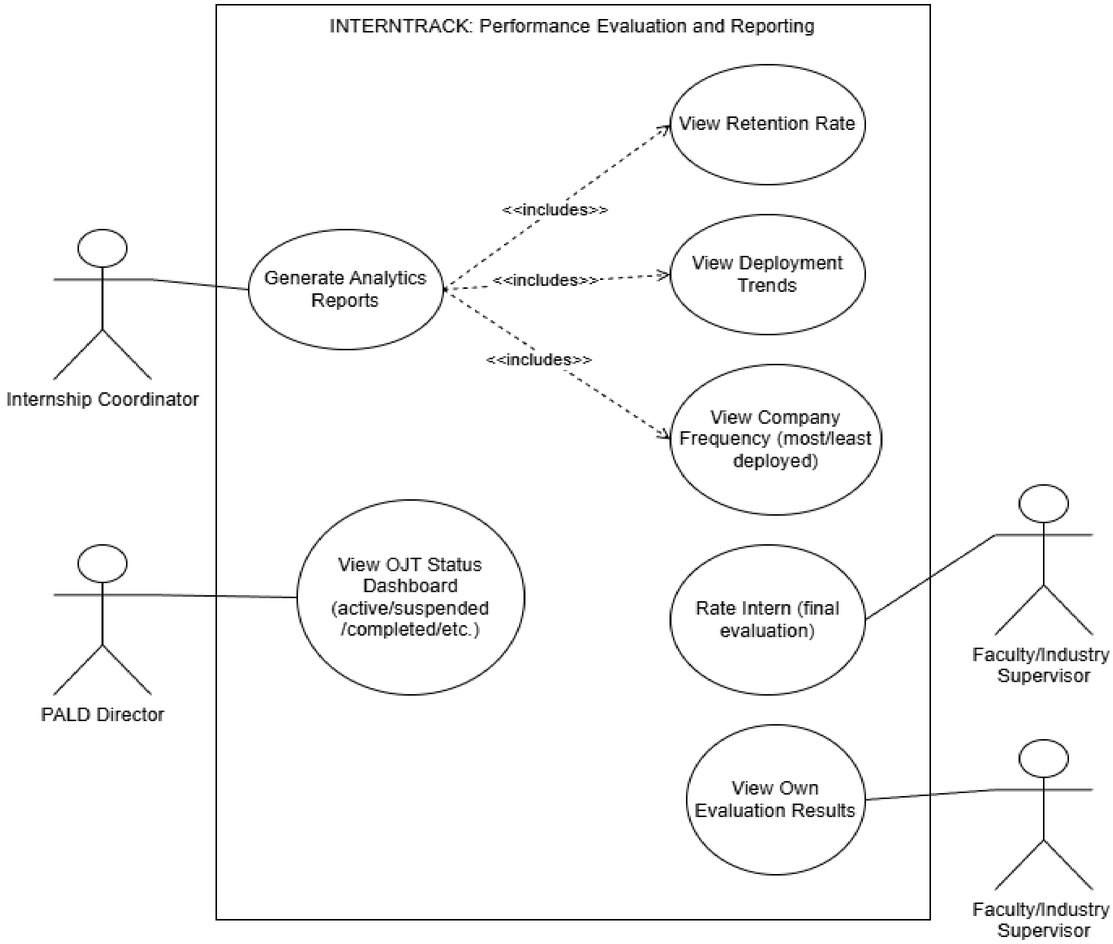

Figure 6. Use Case Diagram for Performance Evaluation and Reporting

This use-case diagram illustrates how the INTERNTRACK system can assist in tracking and supervising intern progress and real-time validation through dashboards. The Internship Coordinator monitors progress, schedules activities and keeps track of at-risk interns; attendance and performance metrics are monitored in real time by the supervisor and PALD Director.

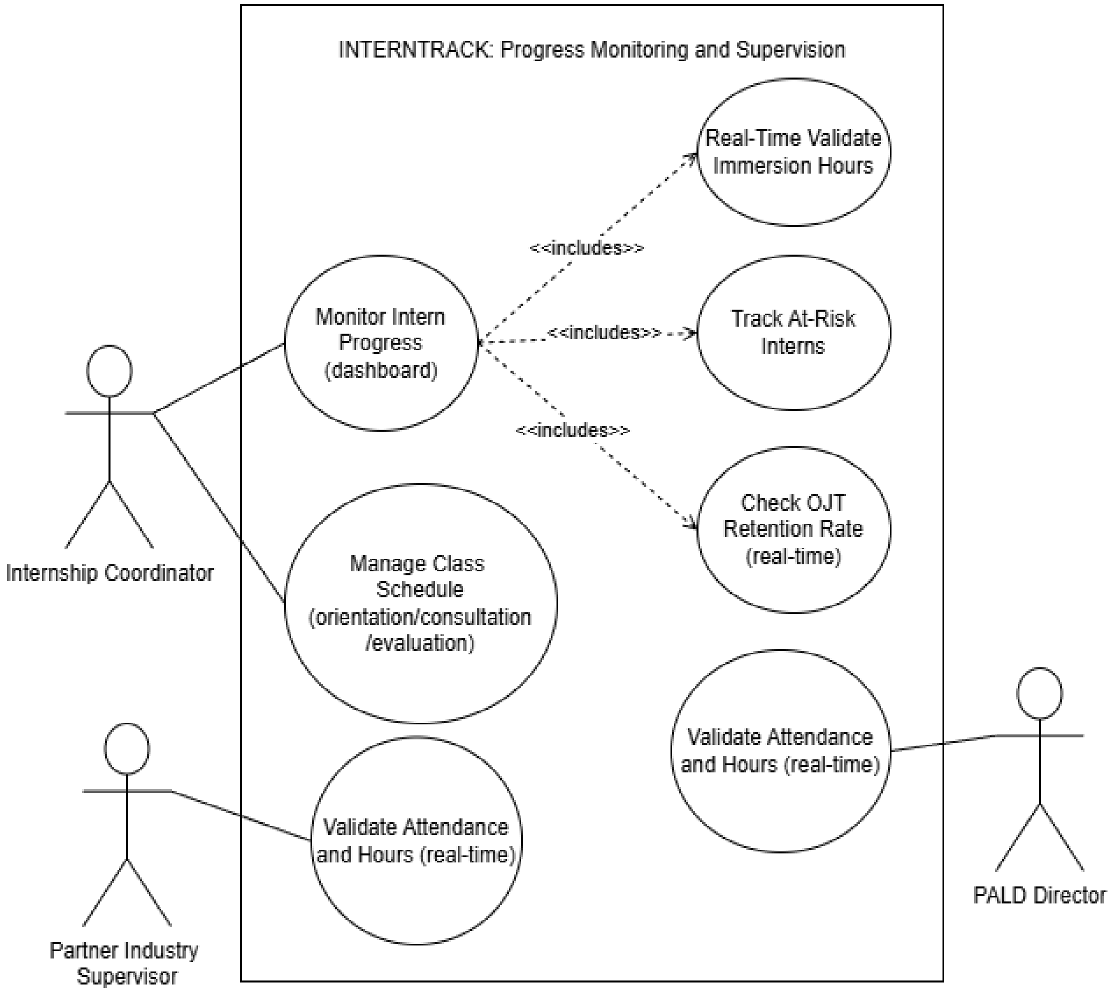

Figure 7. Use Case Diagram for Progress Monitoring and Supervision

Activity Diagram

This chapter describes the internship management processes and presents them in the form of activity diagrams implemented in INTERNTRACK. Through the mapping of actions, decisions, and user responsibilities, these processes are handled in student records, placements, submitting papers to journals, ensuring documents are compliant, progress monitoring, and evaluating performance. The diagrams reveal that there is a lack of any manual operations in the management process, which is defined by a set of rules, each of which is associated with a specific condition. Therefore the activity diagrams show how efficient, organized and effective you can manage your internships using the INTERNTRACK.

This diagram outlines a process for the student to submit their personal and documentation information for review to be included in a student profile for internships. The profile is reviewed by the Internship Coordinator who either returns it for modification or submits to appropriate academic staff for approval.

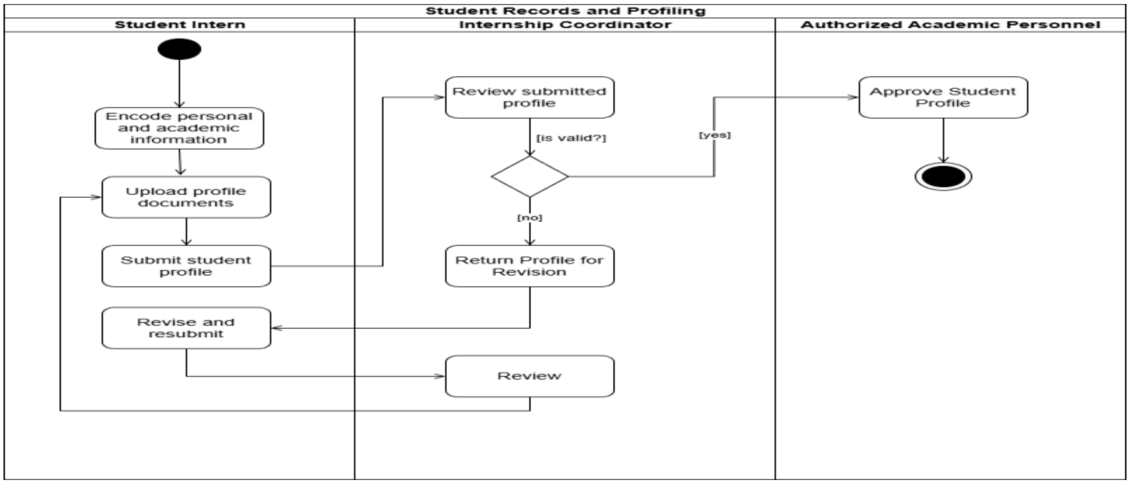

Figure 8 Activity Diagram for Student Records and Profiling

The placement and deployment monitoring process includes an Internship Coordinator, Student Intern, and Faculty Supervisor to review internship records, ensure fulfilment of requirements, and approve placements. It also provides examples of the management of alerts, placement status and faculty assignments by the three roles to ensure coordination.

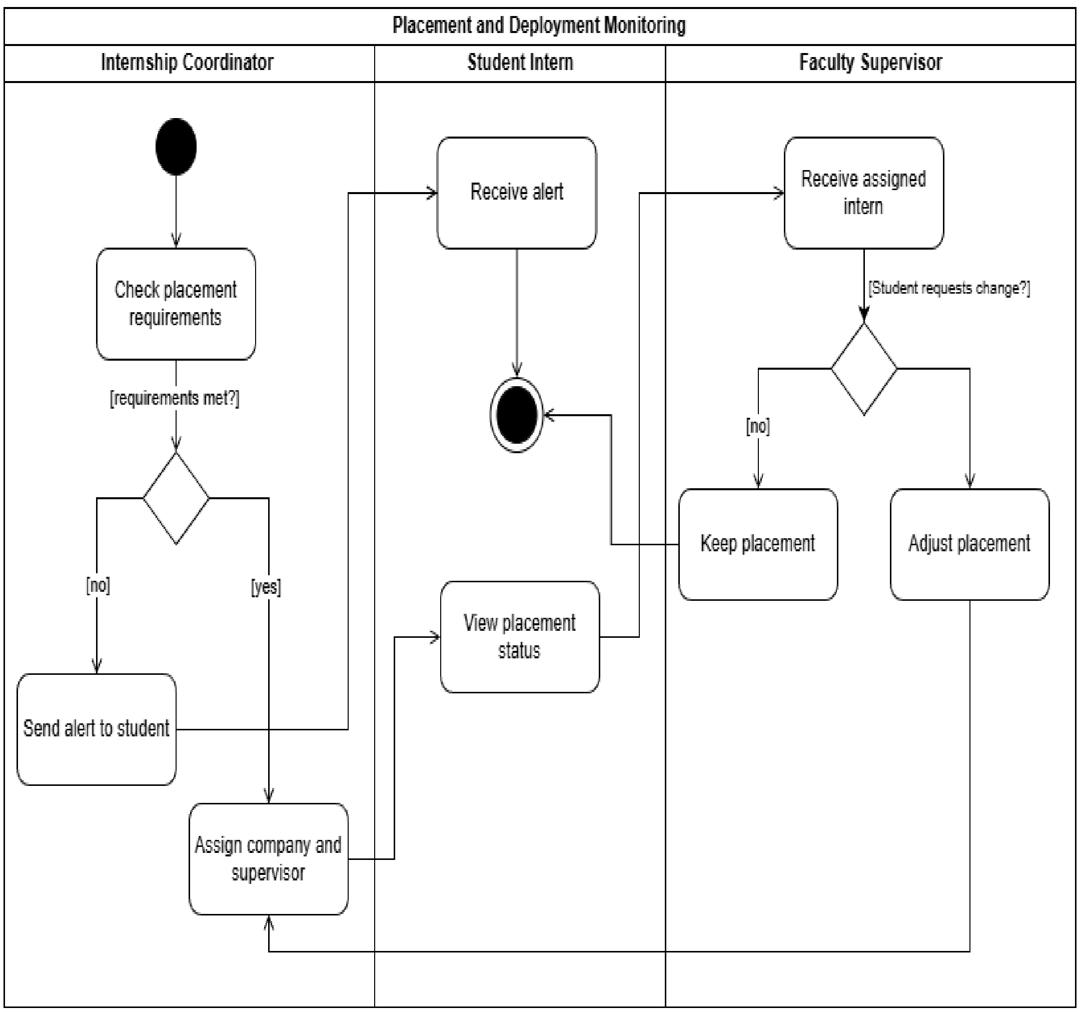

Figure 9 Activity Diagram for Placement and Deployment Monitoring

The diagram shows the digital journal submission process: a student intern submits a digital journal to DTR and hours which are then reviewed by a faculty supervisor. If the journal is accepted it is approved, otherwise it is returned to the student for revision and resubmission.

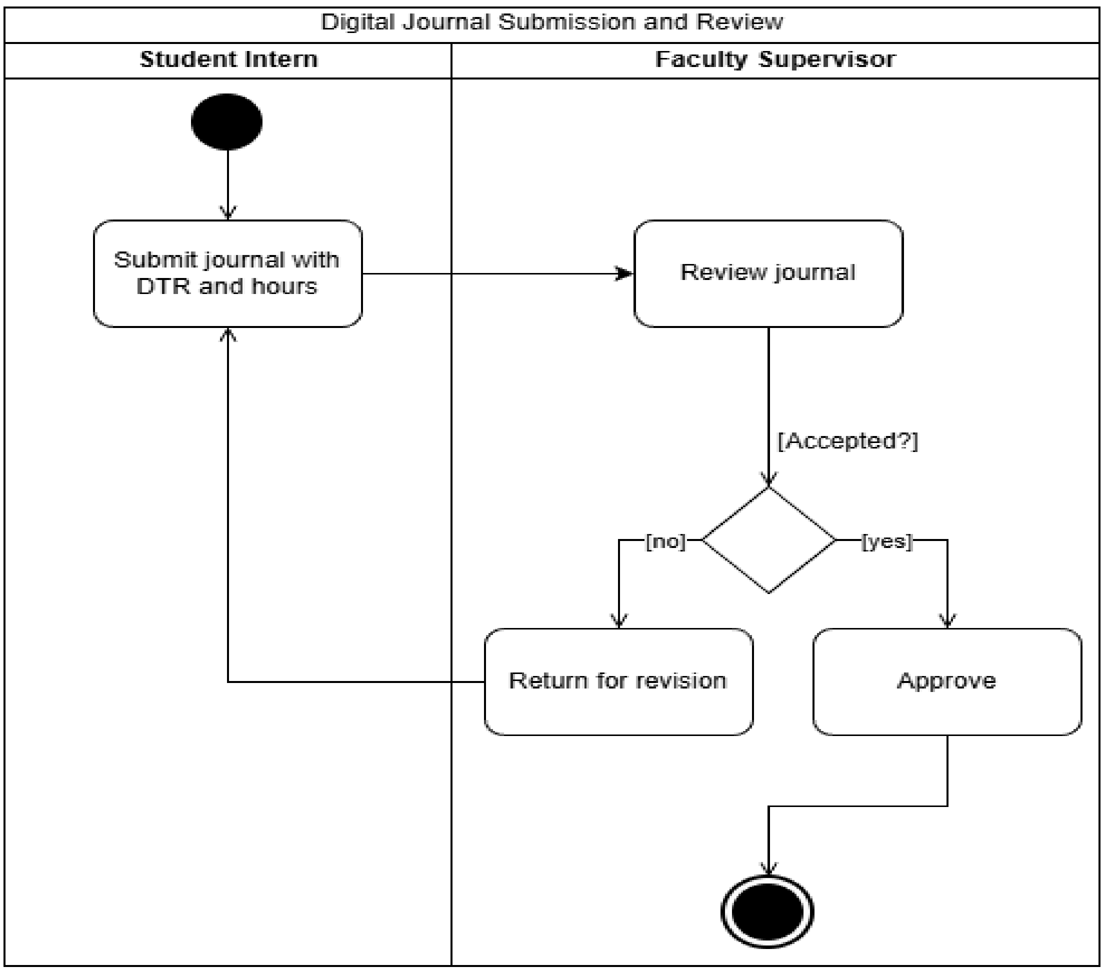

Figure 10 Activity Diagram for Digital Journal Submission and Review

The document compliance workflow is illustrated in the diagram, in which a student intern submits a document, which is then reviewed by the internship coordinator to determine if it is complete and correct. If valid, it is submitted to the appropriate academic staff member for approval and compliance is updated, if not the student will have to re-upload and check status.

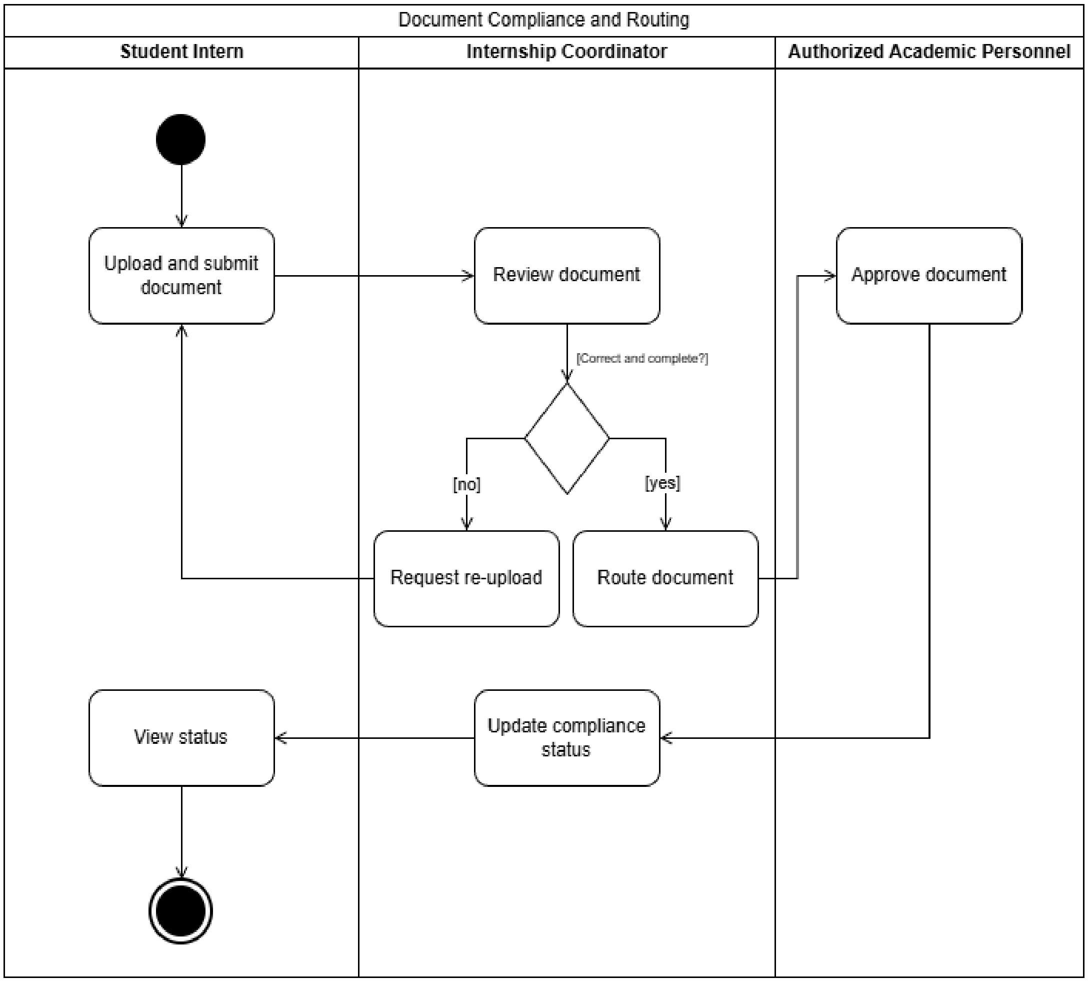

Figure 11 Activity Diagram for Document Compliance and Routing

The diagram illustrates a workflow of monitoring intern performance where four roles are involved: Faculty Supervisor, Internship Coordinator, Student Intern and PALD Director. Identified issues by the supervisor send out notification and tracking by the coordinator, and continuous monitoring and analytics maintain communication among both the intern and director.

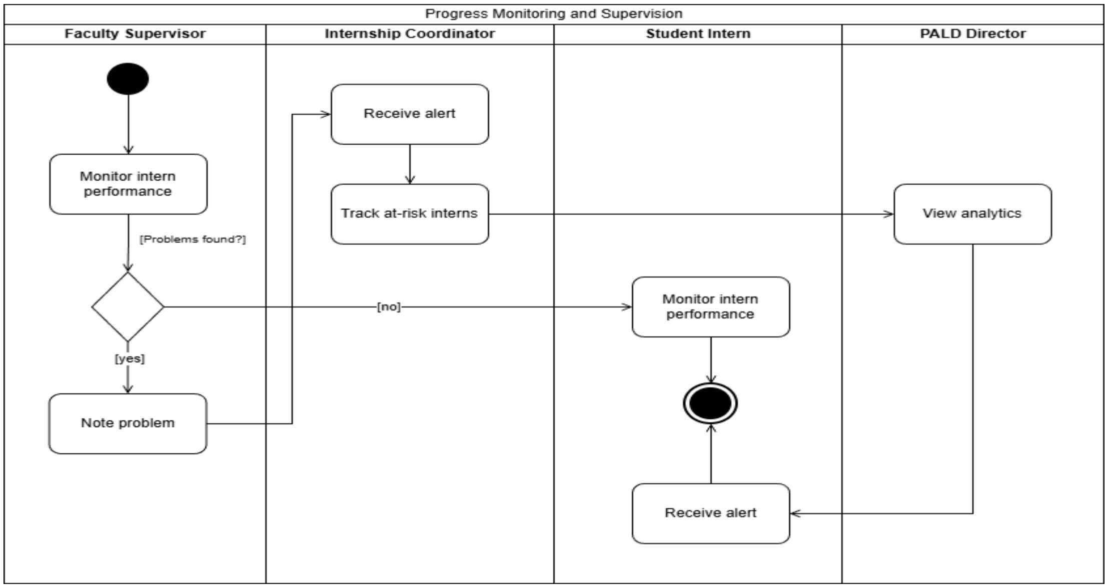

Figure 12 Activity Diagram for Progress Monitoring and Supervision

The graph below summarizes the evaluation process in which the industry supervisor rates the intern and the internship coordinator makes sure that all the evaluations are filled out before creating the report. The report is considered by academic staff for approval/revision and then shared with the student intern.

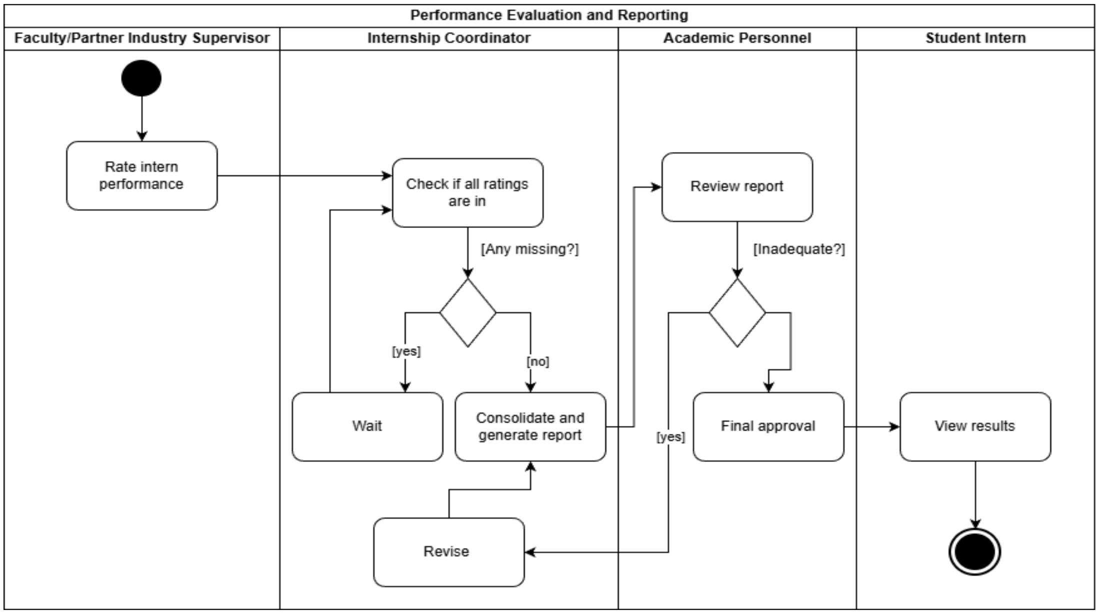

Figure 13 Activity Diagram for Performance Evaluation and Reporting

Sequence Diagram

The INTERNTRACK sequence diagrams depict the time-ordered communication between users, interfaces, controllers and the database, including how requests are made, processed, validated, stored, retrieved, and then the response is sent back. They offer a more detailed representation that shows more than just message flows as use case diagrams do; they show the behavior of the systems involved in each step of the process. These diagrams are part of the third section of the paper, focusing on data flow, processing logic and coordination between the modules, showing how the major components of INTERNTRACK are integrated.

The sequence diagram illustrates the flow of events during the creation of a profile, including the student intern logging in, providing their personal and academic details, the profile controller validating the data, saving the profile and informing the coordinator and student. It emphasizes the role of key players, error management (including duplicate profiles), and that only valid profiles move forward, as problems at this point can affect all future internship activities.

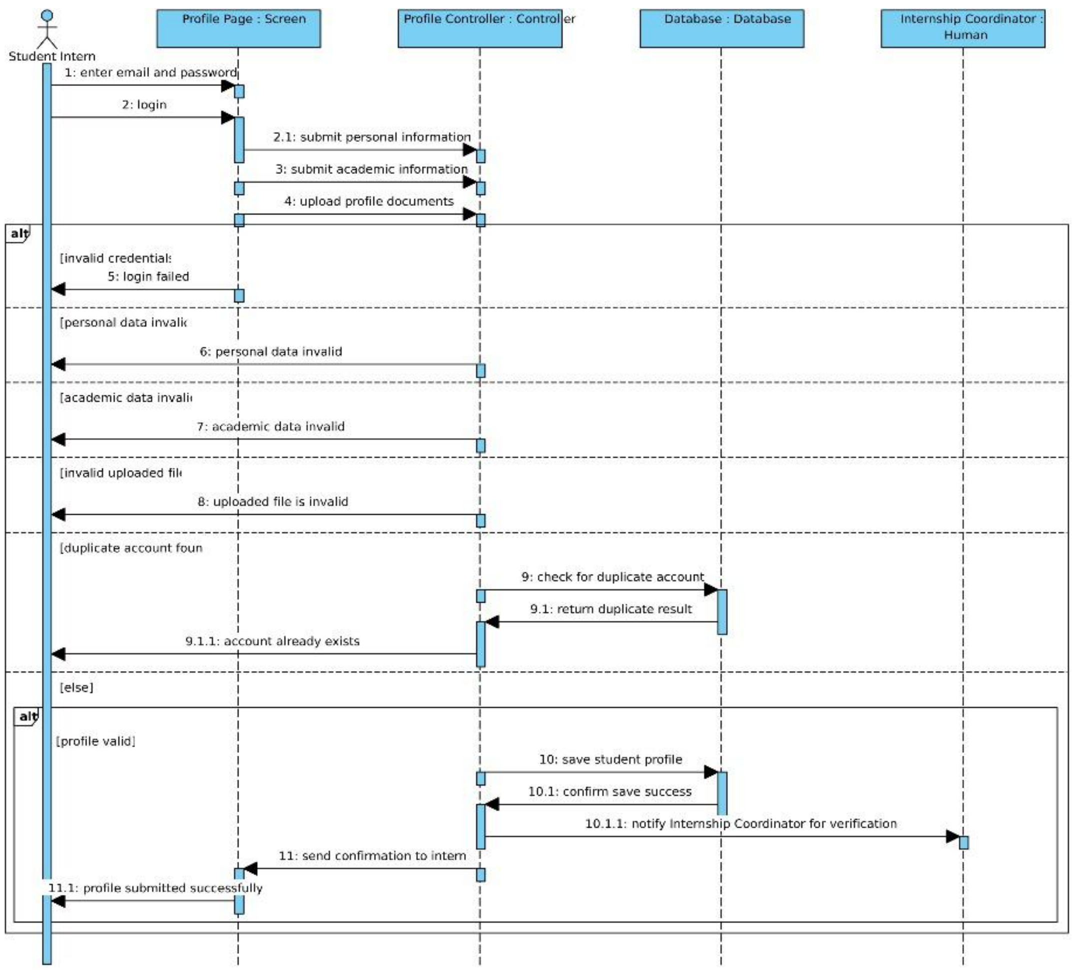

Figure 14. Sequence Diagram for Student Records and Profiling

The sequence diagram illustrates the process of assigning internship to a student, where the coordinator chooses a student, a company and a supervisor, the system checks the eligibility and availability of the slot, updates the database and informs the student. It identifies important decision points, alternative results (ineligibility or unavailable slots), and provides a controlled and traceable workflow to guarantee accurate placement management.

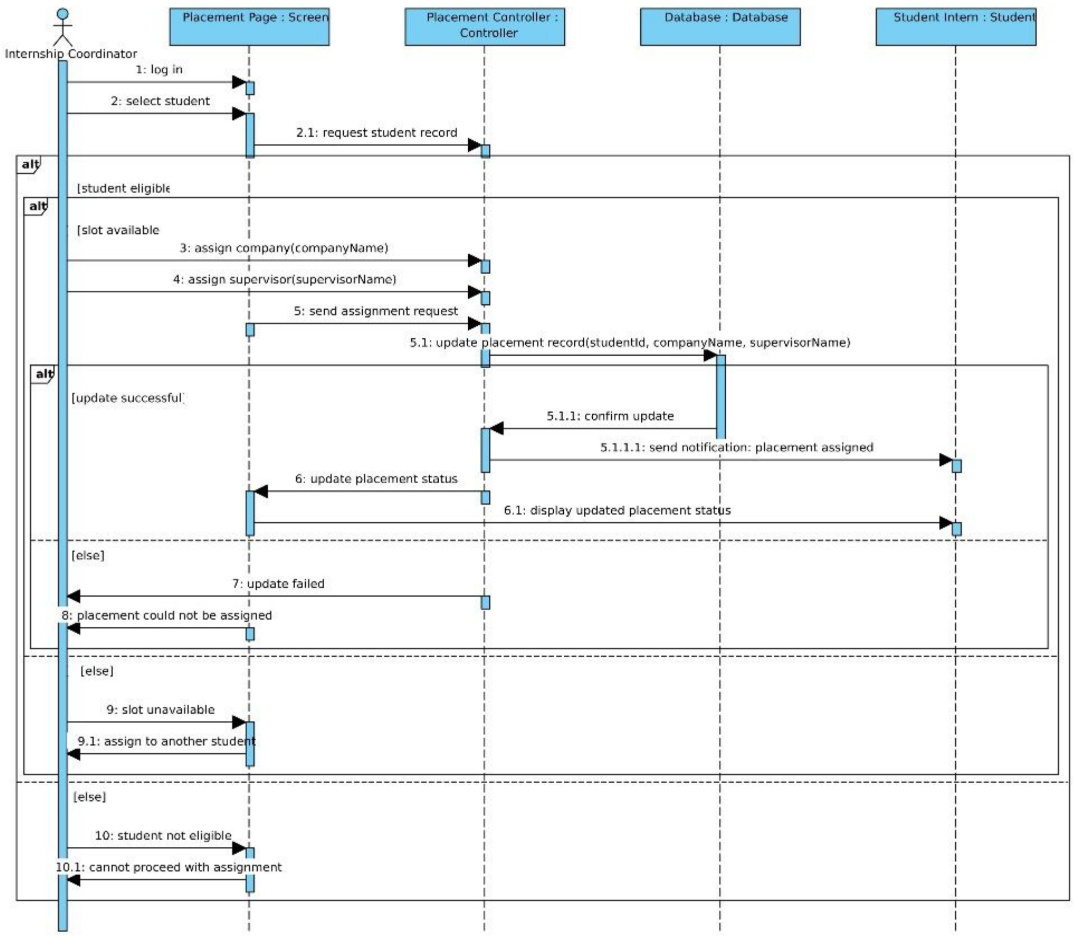

Figure 15. Sequence Diagram for Internship Placement and Deployment Monitoring

The sequence diagram illustrates the process of submission, validation, and storage of journal entries, followed by a review by the faculty supervisor for approval, rejections or pending state. It identifies important actors and possibilities of alternative outcomes, and provides a traceable workflow with good documentation, verification and feedback.

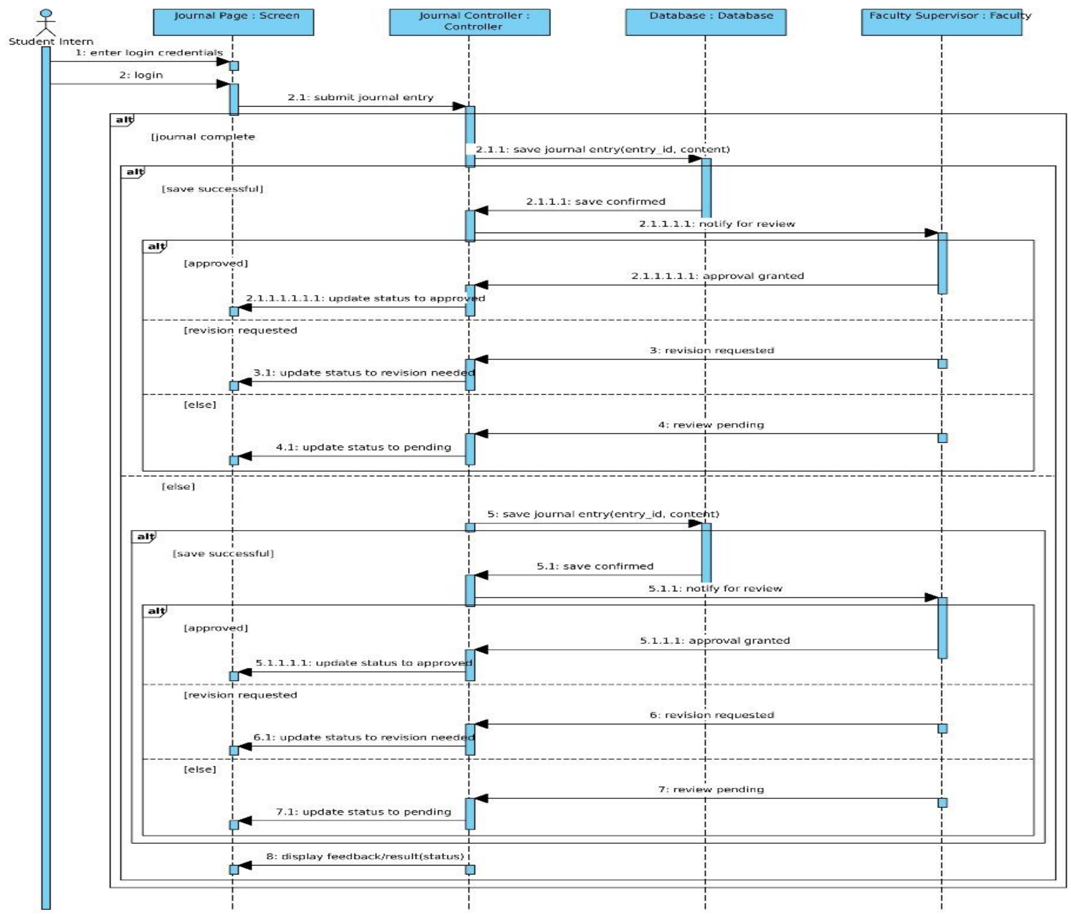

Figure 16. Sequence Diagram for Digital Journal or Logbook Submission and Review

The sequence diagram is an illustration of the time-ordered process of submitting a requirement, which the system validates and stores, then the internship coordinator reviews and approves, rejects, or requests revisions, and updates the status and informs the student. It identifies critical actors and alternative routes (valid/invalid submission, approval/revision), creating a traceable and feedback oriented compliance process.

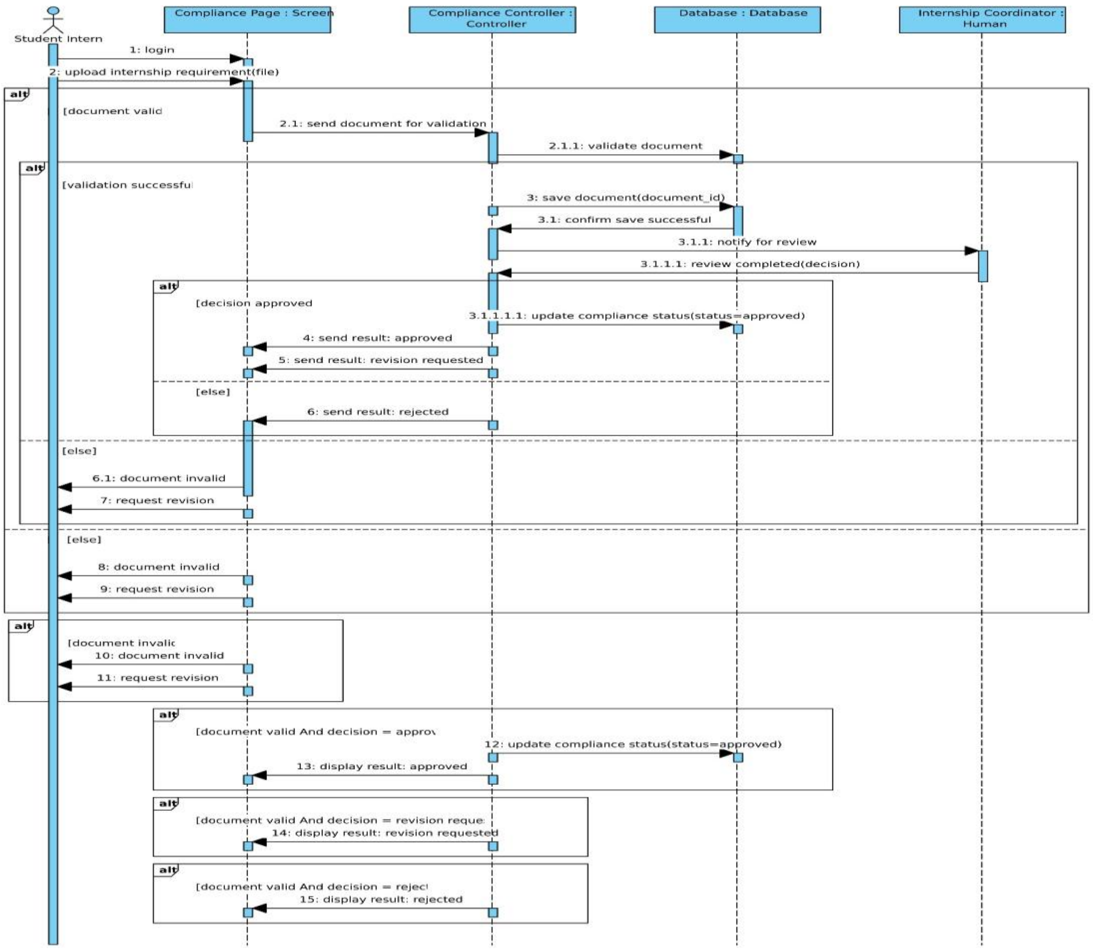

Figure 17. Sequence Diagram for Document Compliance and Routing

The sequence diagram demonstrates the process of progress monitoring, where the progress data of a student is retrieved by the faculty supervisor, who will review the summary generated, and will enter supervisory remarks in the student's progress data, which will be stored in the database for the student to retrieve. It emphasizes critical actors and other conditions (including delays and missing requirements), and ensures a structured and data-driven process for monitoring performance and giving real-time feedback.

Figure 18. Sequence Diagram for Progress Monitoring and Supervision

The sequence diagram illustrates the time-ordered events of submission of an evaluation, input of ratings and comments by the industry supervisor, validation and storage of the evaluation, generation of the evaluation report for the internship coordinator, and confirmation of evaluation submission. It identifies the key players and what may go wrong (e.g. only part of the form is filled in, or a child does not save it), and provides a clear and consistent flow of work for evaluation and reporting.

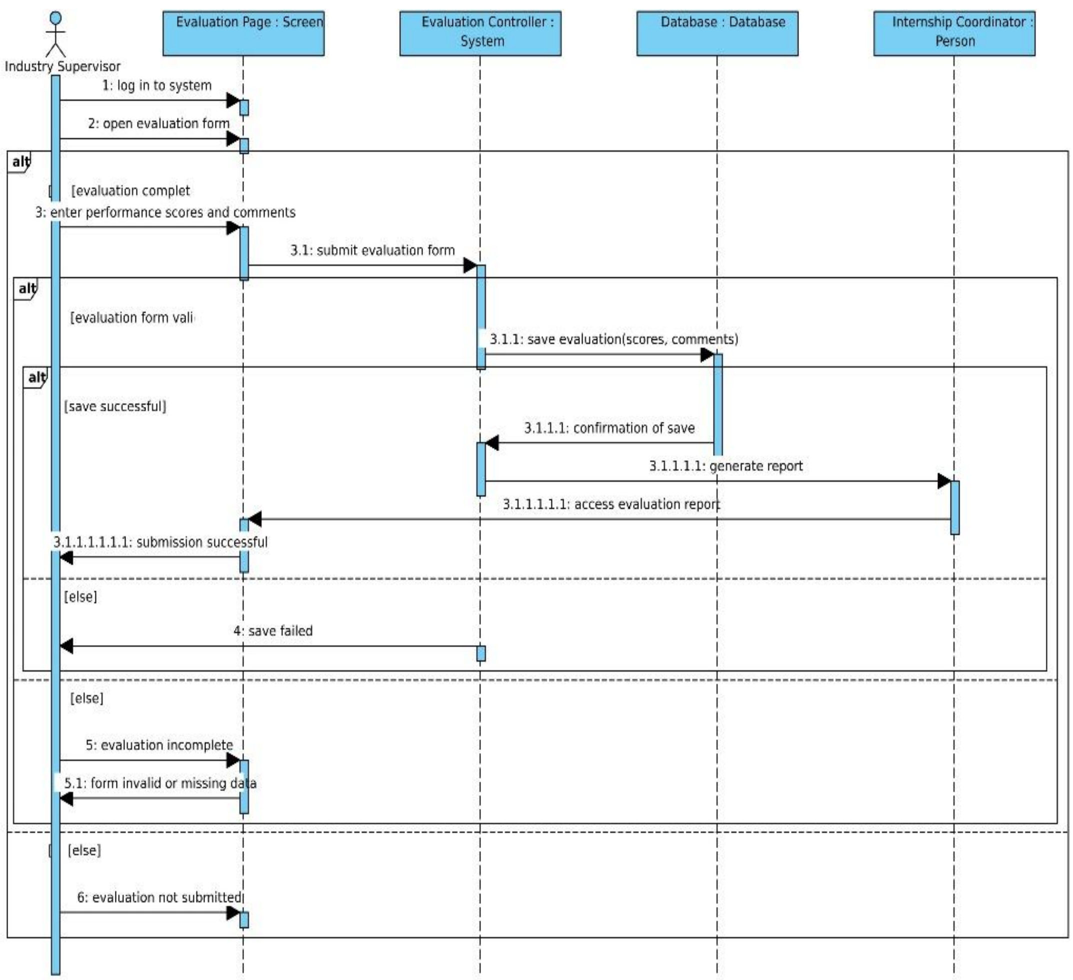

Figure 19. Sequence Diagram for Performance Evaluation and Reporting

Data Flow Diagram

The INTERNTRACK data flow diagrams present the data flow that occurs in the proposed internship management system. They are added in Chapter III to illustrate the way the information is handled and converted as it flows through the different functional modules of the system, showing the relationship between the external entities, major system operations and data repositories. The research explains the general behavior of the movement of data in the profile data, placement data, journal record data, compliance data, monitoring information, and evaluation outcome data using the context diagram and the level 1 DFD in an organized and traceable manner.

The context diagram for INTERNTRACK shows the whole system as a single process, which is connected to five external entities: Student Intern, Internship Coordinator, Faculty Supervisor, Partner Industry Supervisor, and Authorized Academic Personnel. It depicts the data exchange between each entity and the system, including things such as profiles, journals, evaluations, placements, and reports, along with processing and output of relevant data, such as feedback, approvals, and summaries. This diagram defines the scope of the system and is the starting point for creating a Level 1 DFD that is used to represent and explain the detailed sub-processes of the system, such as profile management, placement, monitoring, compliance, and evaluation.

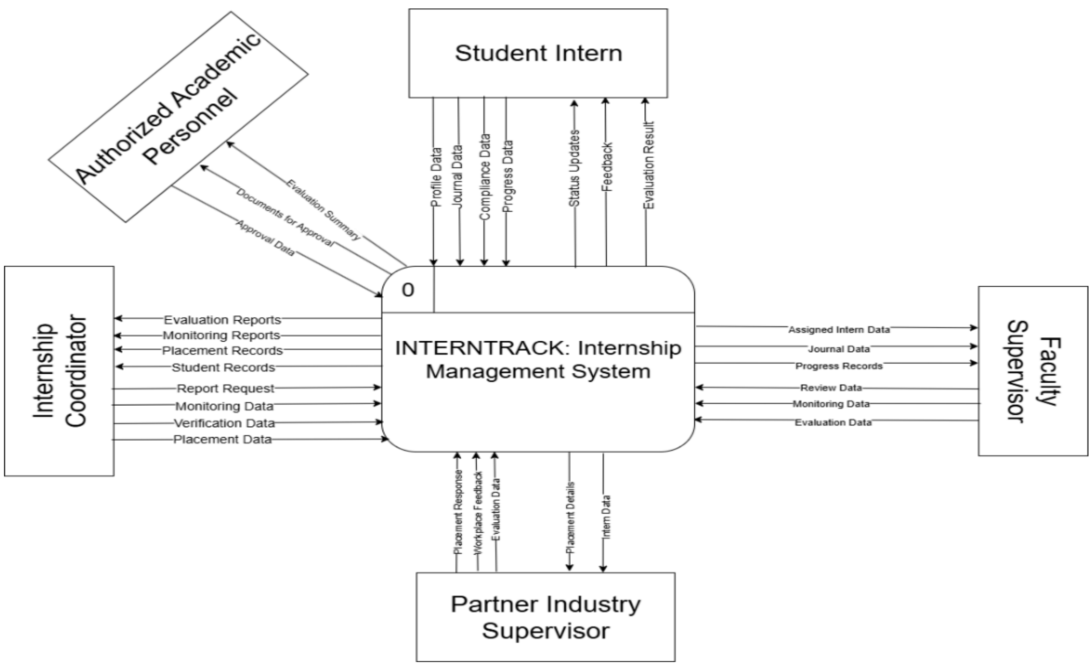

Figure 20. Data Flow Diagram Level 0 for INTERNTRACK: Internship Management System

The diagram demonstrates the flow of profile data, placement requests and journal entries into key processes that create and maintain related profile, placement and journal records throughout the internship process: placement management, digital journalism, compliance handling, monitoring and evaluation. It also emphasizes the responsibilities of supervisors and academic staff in making initial authorization and evaluation requests, and of the system to ensure they are dealt with and approved at each stage.

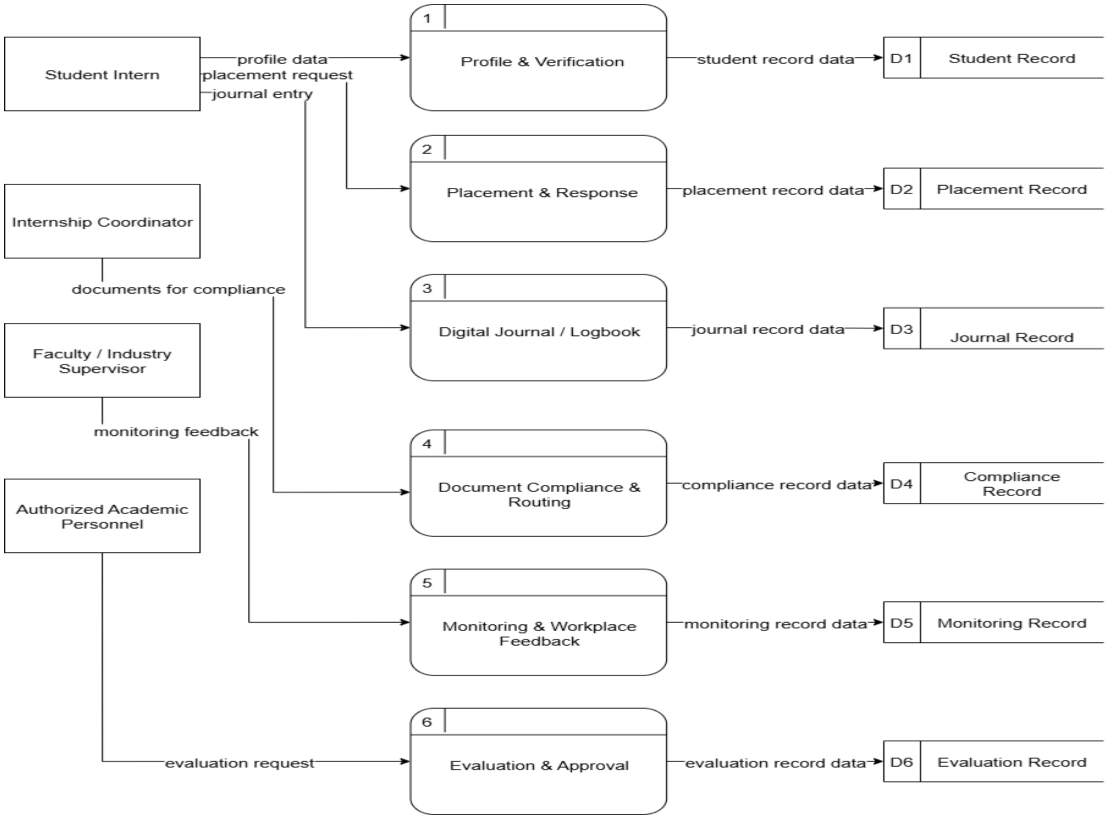

Figure 20. Data Flow Diagram Level 1 for INTERNTRACK: Internship Management System

System Architecture Diagram

The system architecture diagram for the INTERNTRACK system shows the end-to-end internship management system architecture, which is divided into various modules for managing user authentication, student records, internship placement tracking, internship journal submission, document compliance, progress monitoring, evaluation, and data persistence. The diagram has a structured flow starting from the initial user input, which includes profile, academic data, and documents, leading to the core processes such as placement, journal review, compliance, and supervision, and culminating in the final evaluation and reporting. Each major function is associated with related data save operations (e.g., saving placement assignments, journal entries, document records, monitoring records, etc.) and illustrates the flow of data from process to persistent store and back to process. This architecture provides role-based access to students, faculty supervisors, compliance officers and authorized academic staff members, providing good separation of concerns and audit trails for internship management.

The diagram shows the concept of an end-to-end system for managing internships, namely, INTERNTRACK, which involves all aspects of managing an internship, including user authentication, profile maintenance, placement, intern journaling, compliance, monitoring, and assessment, and is interconnected and joined together by a series of processes and records. It demonstrates the movement of student data, submission and feedback within the different modules that provide reports, decisions and status updates and that were validated and approved by supervisors and academic staff. Overall, it provides a representation of a systemized method in which all internship activities are recorded, managed and archived for ongoing monitoring and final assessment.

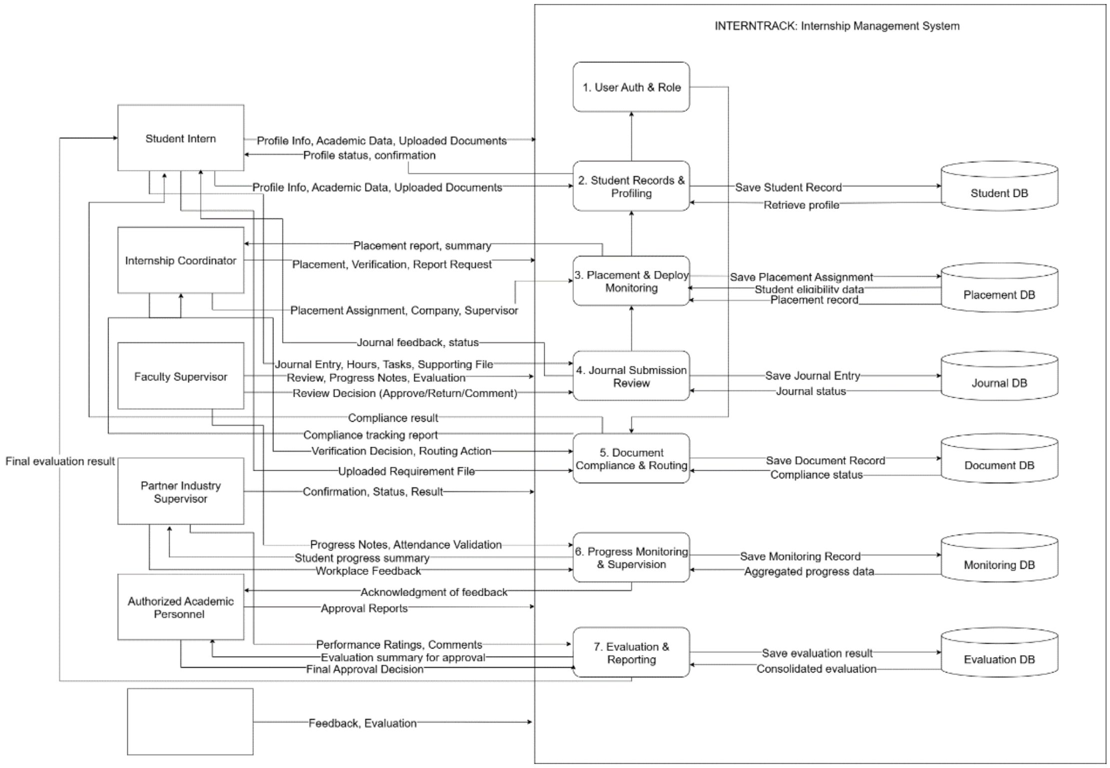

Figure 22. System Architecture Diagram for INTERNTRACK: Internship Management System

Storyboarding

Storyboarding for INTERNTRACK is visualizing the interactions between the user and the system at important internship management phases. Every storyboard panel illustrates a specific user's activity, like a student submitting a journal entry, a faculty supervisor checking progress, a faculty member approving a placement, etc., and how the system reacts and moves with data. It can help map out user journeys, pinpoint areas for improvement, and make sure that each step of the process from the registration to the final evaluation is tailored to the user's expectations. The storyboard will help you design for validation and communicate with stakeholders by showing examples of such scenarios as document compliance routing, monitoring feedback, and evaluating approvals.

The student intern starts by logging in to INTERNTRACK, with her student ID and password, or registering as a new student using her university credentials. The system verifies the information, registers the student and takes him/her to a personalized dashboard. On the dashboard, the student can see the pending tasks (upload MOL document) and hour tracking (280 hours and 14 minutes), journal count, and other document alerts. The student then clicks on the My Profile section, provides academic details, uploads necessary paperwork and submits the profile for coordinator consideration and approval. The system now notifies the profile as pending review and updates the dashboard.

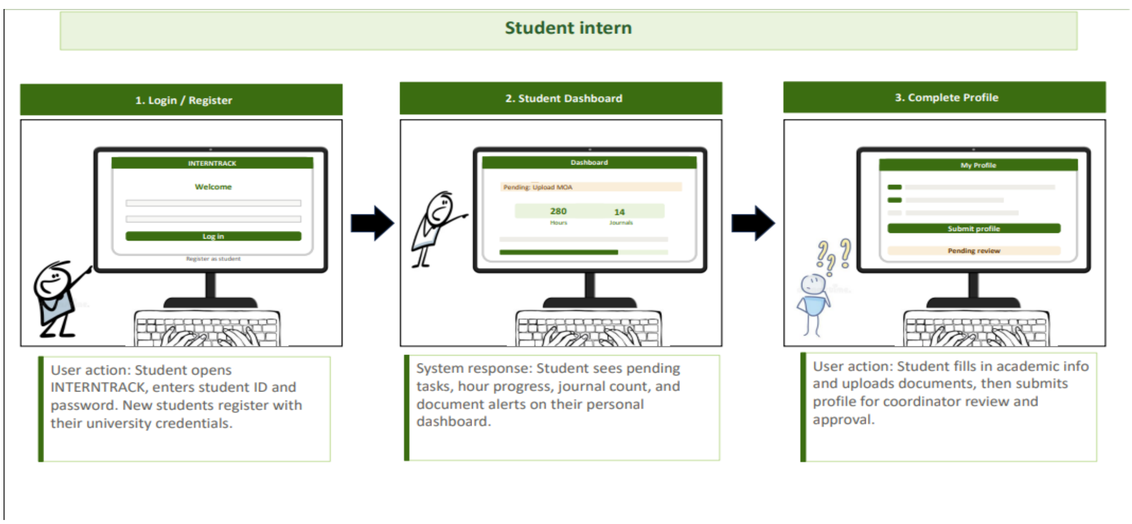

Figure 23. Storyboarding for INTERNTRACK: Internship Management System

The student continues their intern journey by checking the placement details on the My Placement page, where they get to know the company they are placed at and the faculty supervisor, and the deployment status changes to Approved. The student then selects a document type and uploads a document, which the system will sent to the coordinator for review; indicating that it has been attached and sent for review. Finally, the student makes a daily note in the journal, noting tasks and hours for a particular day (Day 42: 8 hours) and the system sends a notification to the faculty supervisor who then reads the student's entry.

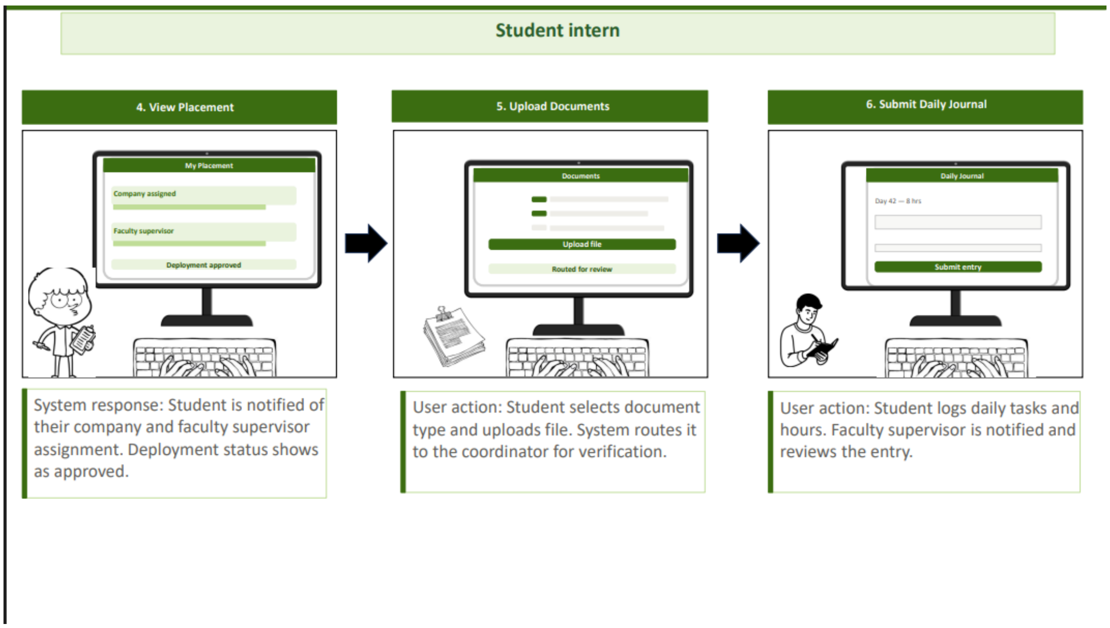

Figure 24. Storyboarding for INTERNTRACK: Internship Management System

Later, the student logs onto the system, and the system shows the student a progress bar (280 of 400 hours), a bar of supervisor feedback (Good progress!), and a status bar (On track!), and alerts are issued if the student is at risk of falling behind schedule. Finally, the student sees the end evaluation section of the system, and the system will display a combined final rating from the faculty and industry supervisors, indicating that the internship has been completed.

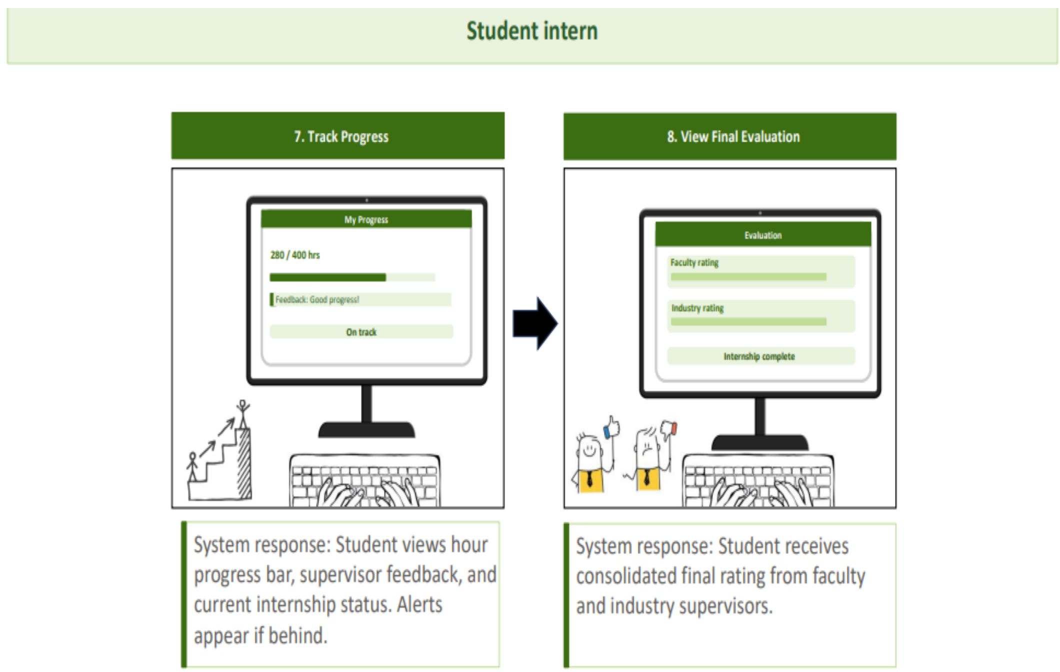

Figure 25. Storyboarding for INTERNTRACK: Internship Management System

The coordinator logs in with his/her admin credentials, and the system automatically gives him/her coordinator-level access as soon as he logs in. Then the coordinator will be presented with a dashboard that displays an overview of all active interns, 3 profiles for review, 42 interims, and all roles. The coordinator can access student profile (Juan Dela Cruz and Maria Santos) on the dashboard, checks the submitted documents and passes them back for revision if required.

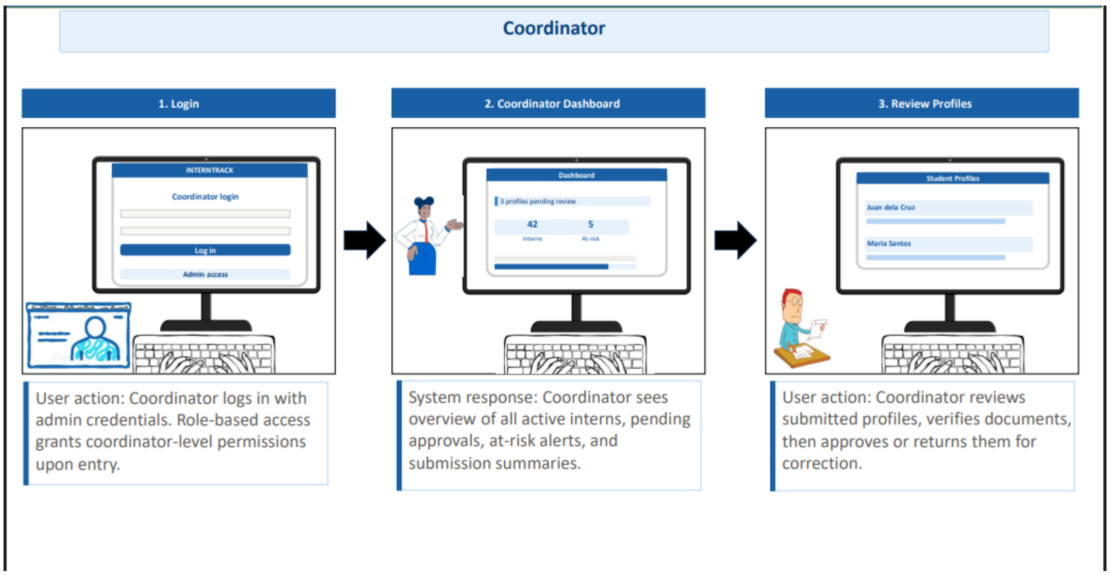

Figure 26. Storyboarding for INTERNTRACK: Internship Management System

The coordinator journey continues with coordinator assignment of placements, verification of requirements, and assignment of faculty supervisor and authorization of placement. The system then marks the placement and changes the student status to placement. Then, the coordinator checks students' documents, like Juan dela Cruz's MQA and Maria Santos's Walter; if they are correct, he/she approves them, and if not, he/she returns them to the student for revision. The coordinator also checks the journal submissions by consulting the journal log, seeing that Juan on Day 42 and Maria on Day 41 are late. System keeps track of all journal submissions and flags late and missing submissions for follow-up.

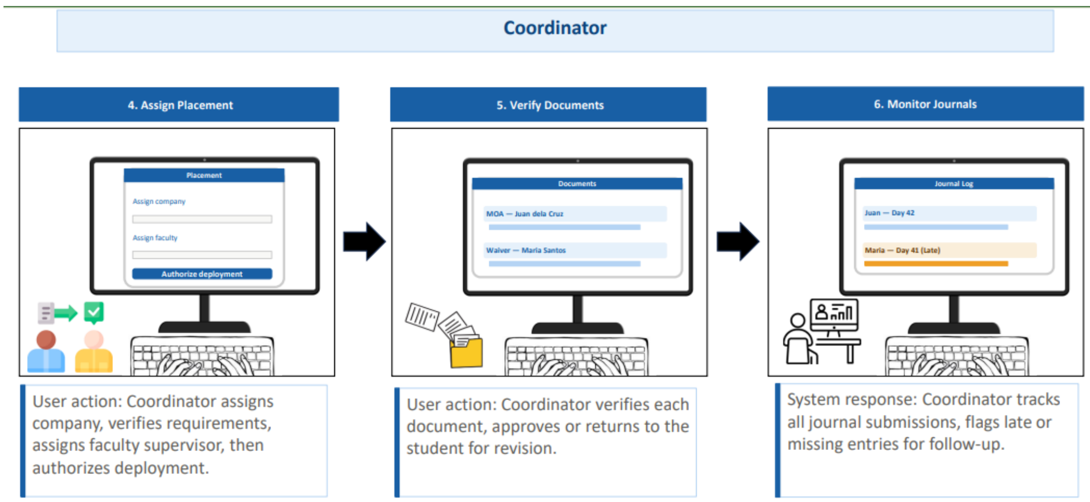

Figure 27. Storyboarding for INTERNTRACK: Internship Management System

Lastly, the coordinator logs into the supervision dashboard to view all the monitoring data, including that of two interns that are at-risk: Juan has 280 hours and Maria has 180 hours (marked at-risk). The coordinator keeps a track of all interns' progress, reviews any alerts for at-risk interns and provides feedback through the platform. Next, the coordinator interprets evaluations by analyzing student records (Juan dela Cruz, Maria Santos, pending), aggregates the ratings, checks for final evaluations and produces internship reports.

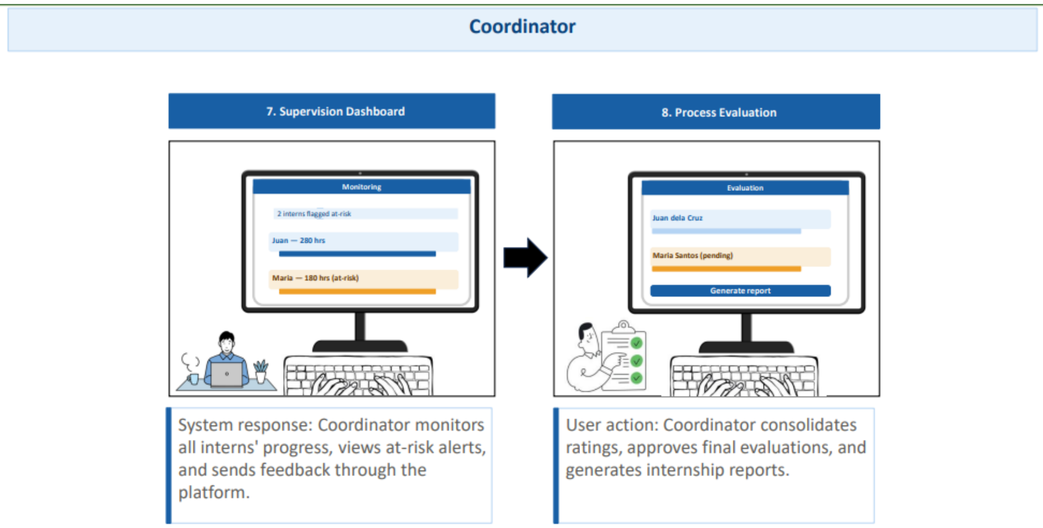

Figure 28. Storyboarding for INTERNTRACK: Internship Management System

Entity Relationship Diagram

The proposed internship management system's database structure is shown in the INTERNTRACK Entity Relationship Diagram, which identifies the main entities, their characteristics, and the connections between them. It is detailed in Chapter III to illustrate the organization, storage and linking of data in the system, which is required for the key modules of profiling, placement, journal management, compliance, monitoring, evaluation and reporting. This figure illustrates the logical structure of the database and how the proposed system maintains the linked, consistent, non-redundant records throughout internships related transactions.

The INTERNTRACK Entity Relationship Diagram outlines the structure of the database system, which includes all of the internship data. All of the internship data is organized and connected in a structured and organized manner. To create relationships between connected tables, it relies on primary and foreign keys, and illustrates interactions between entities like students, companies, and supervisors, such as placements, journal reviews, and compliance tracking. In summary, the ERD ensures seamless data handling, accurate reporting, and sound decision-making across the internship journey.

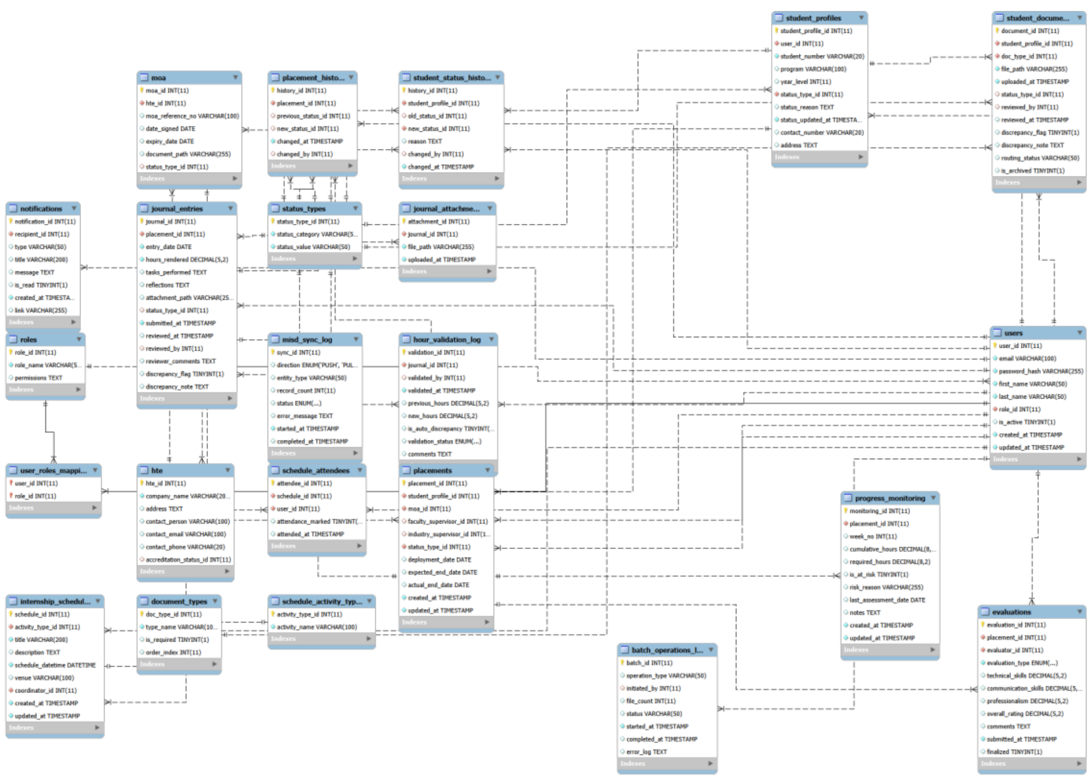

Figure 29. Entity Relationship Diagram for INTERNTRACK: Internship Management System

UI/UX

This document provides explanations of the UI/UX design for the key interfaces of INTERNTRACK in a uniform and consistent format. This is a structured format of explanation used in discussing each screen, ensuring the presentation of the wireframes is organized, thesis-oriented and corresponds with the study's purpose: to create an interface for the internship management system that is user-centered and structured.

The explanations are based on the purpose of the interface and its major components, user interaction that will be expected and its contribution to the internship workflow of the University of Cabuyao. This format can be included in the system development methodology discussion or UI/UX design discussion of Chapter III as the system outputs of the interface.

Each interface is described with the following headings: Purpose of the Interface, Major Interface Elements, User Experience Explanation, and Alignment to the Objectives of the Study for consistency. This consolidated presentation demonstrates the consistency of the interfaces, but also the functional clarity, ease of navigation and role-based usability.

The login interface is the secure and role based entry point of INTERNTRACK, where the user is able to enter his/her workspace by accessing a clearly structured and simple login screen, which contains institutional branding, credential fields as well as important security elements. It creates a user experience that is more usable and trustworthy by helping users like the student intern, coordinator, supervisor, industry partner, academics etc. choose the right module they are looking for without cluttering up the interface and without being overwhelmed by it.

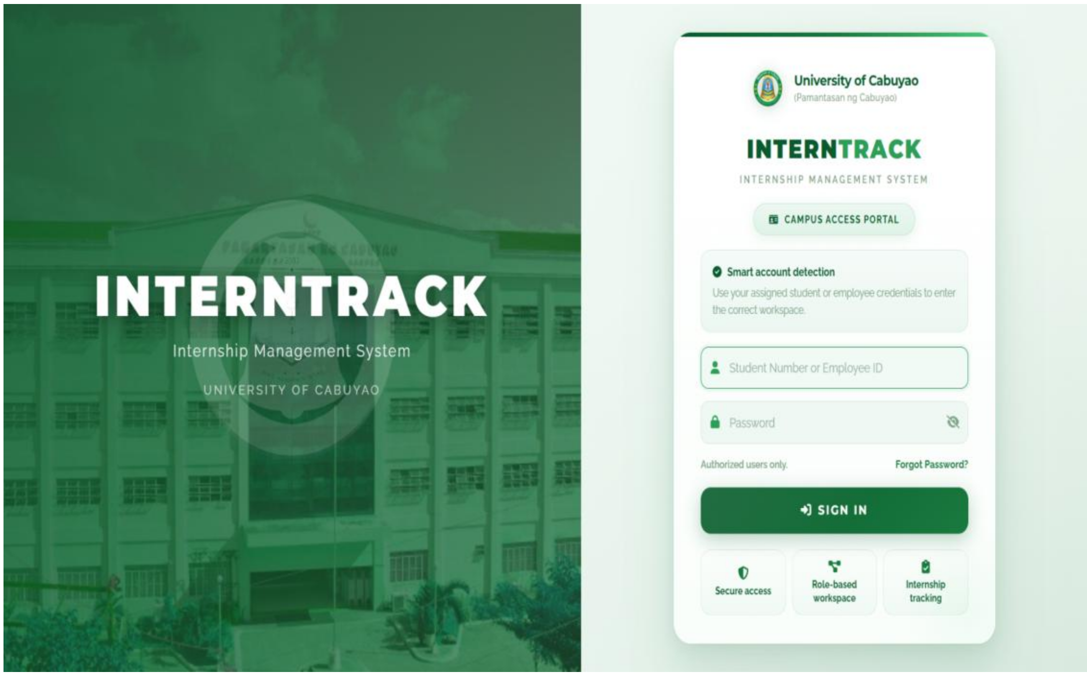

Figure 30. Login Interface of INTERNTRACK

The student dashboard in INTERNTRACK is a central overview page which gives interns a clear overview of their internship status, internship requirements, and internship progress including hours, attendance, journal entries, documents, and announcements, all in one place. It provides a set of structured elements like summary cards, progress bars, activity logs, monitoring widgets, and more to allow the students to immediately see their performance and compliance status. The design also improves the usability and efficiency of the platform, making it easier to navigate and providing up-to-date information about the internships. The platform allows for easy visualization of internships progress and real-time updates, making it more user-friendly and efficient.

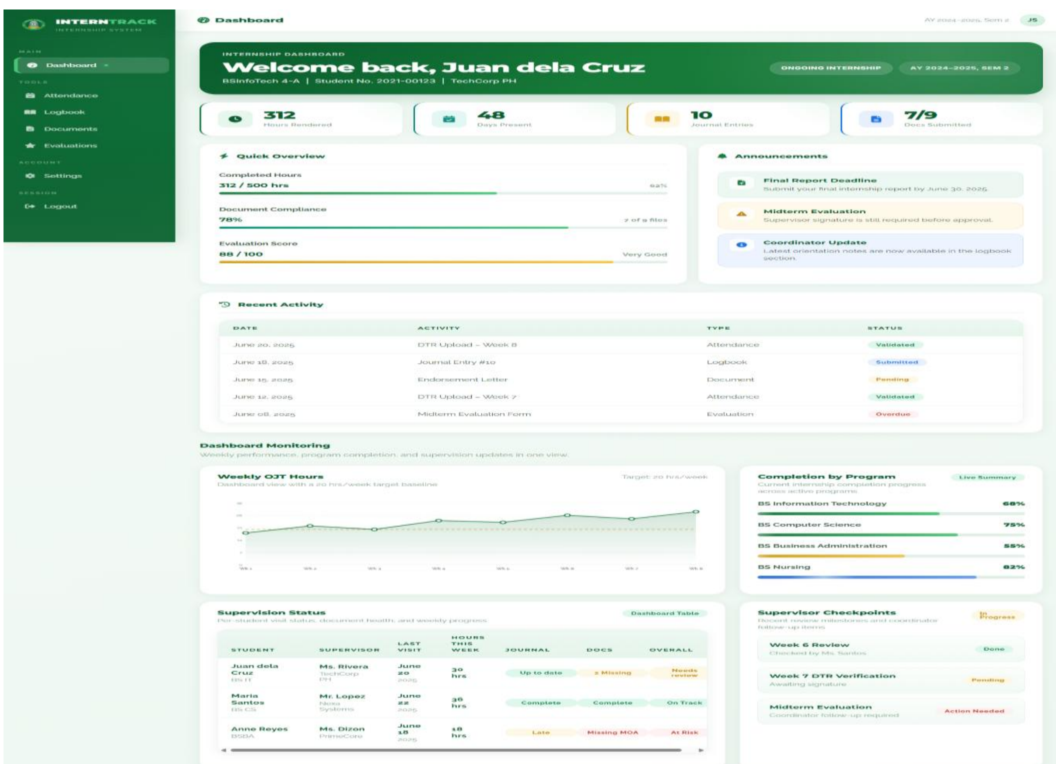

Figure 31. Student Dashboard Interface

The student attendance interface in INTERNTRACK allows interns to submit and monitor Daily Time Record (DTR) files and rendered hours through a centralized module that includes upload fields, date coverage inputs, supervisor notes, and attendance summaries. It is structured with components such as progress bars, status cards, submission history, and checklist panels to help students track their attendance and ensure completeness. Overall, the design improves usability and compliance by organizing attendance tasks clearly, reducing errors, and supporting real-time monitoring throughout the internship period.

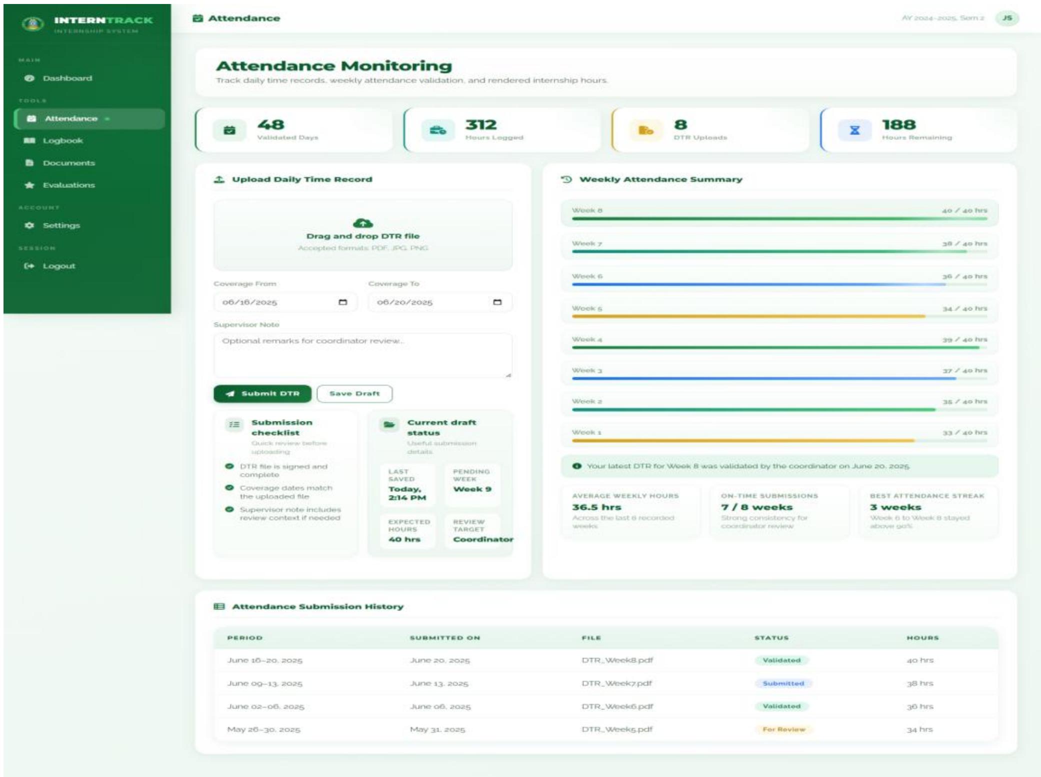

Figure 32. Student Attendance Monitoring Interface

INTERNTRACK's student logbook interface features a structured digital logbook system that automates the manual logging process for interns, enabling them to record daily activities, rendered hours, completed tasks and reflections. It has entry forms, summary cards, and a timeline table with past entries and review status so it's easy to keep track of and refer back to. In conclusion, the interface enhances usability, facilitates efficient journal submission, and provides clarity on progress to both students and supervisors, enhancing overall efficiency and consistency.

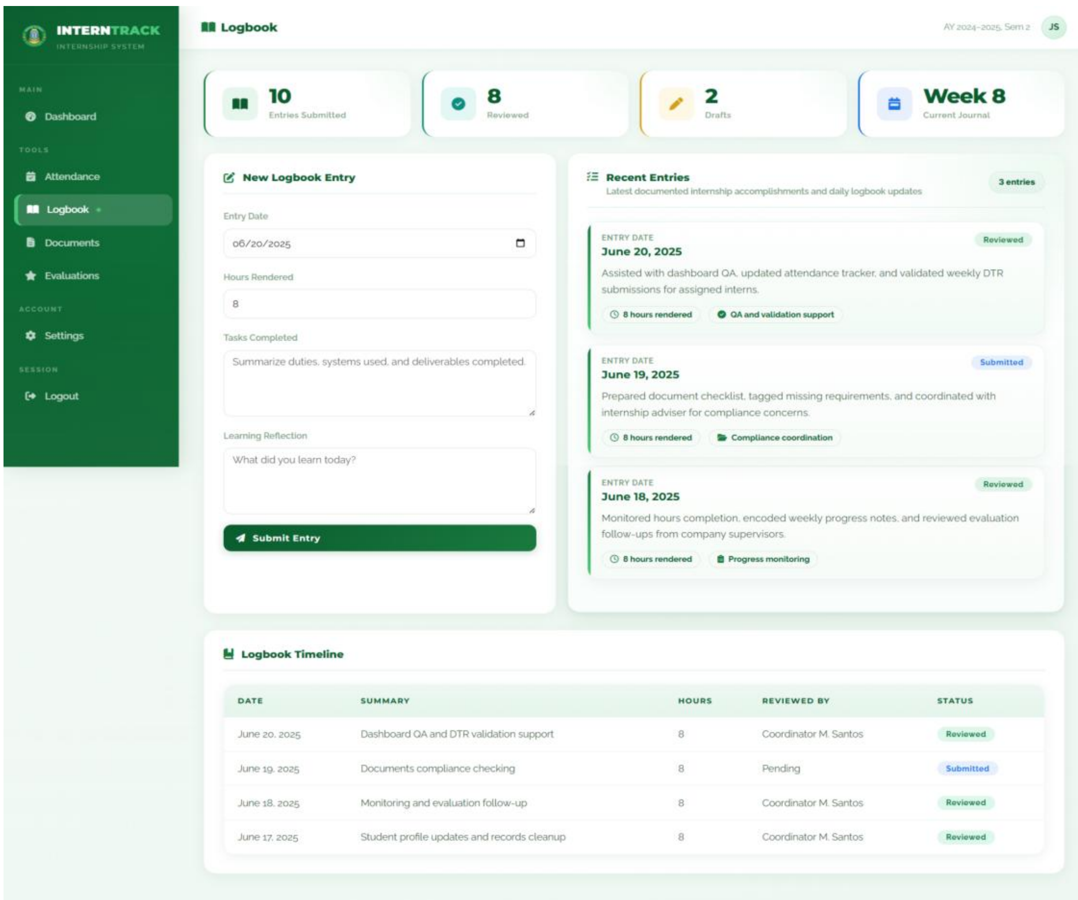

Figure 33. Student Logbook Interface

With the document compliance interface in INTERNTRACK, student interns are able to oversee internship requirements, track missing documents, upload supporting documents, and monitor the status of document submission in a single module. It's packed with features like a requirements table, an upload panel, action buttons, and status summary cards to help manage documents and ensure they're organized. Overall, the interface improves usability and ensures proper compliance monitoring through visibility of the status, submission ease and less confusion in compliance management of internship documents.

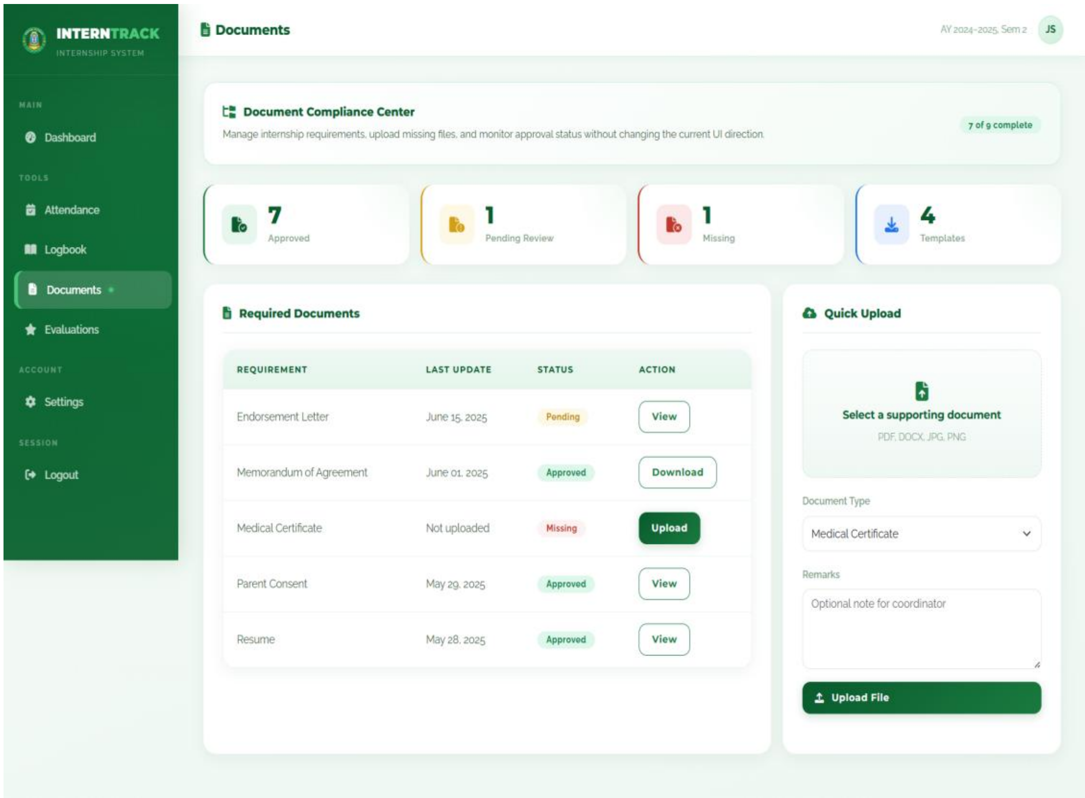

Figure 35. Student Document Compliance Interface

Student interns can keep tabs on evaluation forms, schedules and performance results all in one organized screen with the evaluations interface in INTERNTRACK. It contains score summary cards, evaluation schedules, breakdown panels, and forms tables that show evaluator information, dates and status updates. In general, the interface is designed to make things clearer and more accountable, it shows students their current assessment progress, provides transparency for assessment and keeps students informed about their assessment needs.

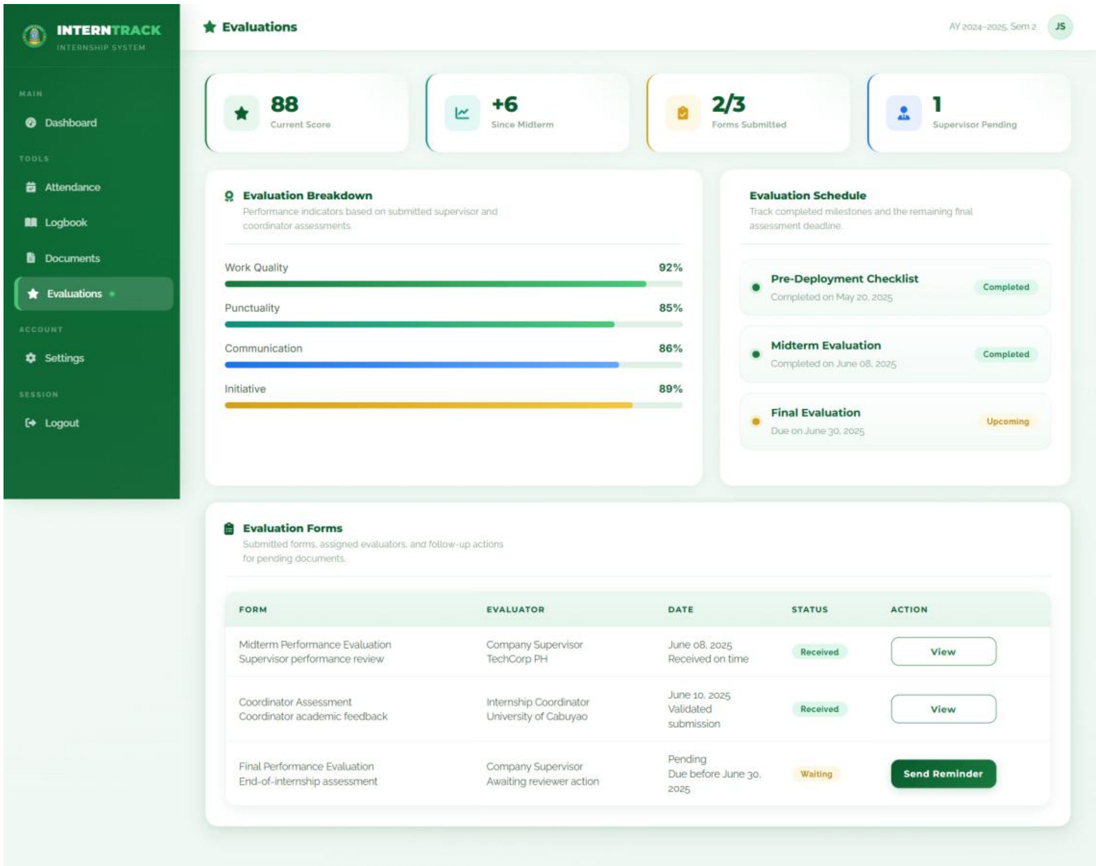

Figure 36. Student Evaluations Interface

The student settings page within INTERNTRACK enables the user to organize and manage their profile information, account security, notification preferences and contact details in one single page. It has features like a profile summary card, setting forms, notification controls, and a security panel for password changes and two-factor authentication. Overall, the user interface offers a cohesive and structured environment for managing personal settings and ensuring accurate account details, while also improving usability and system security.

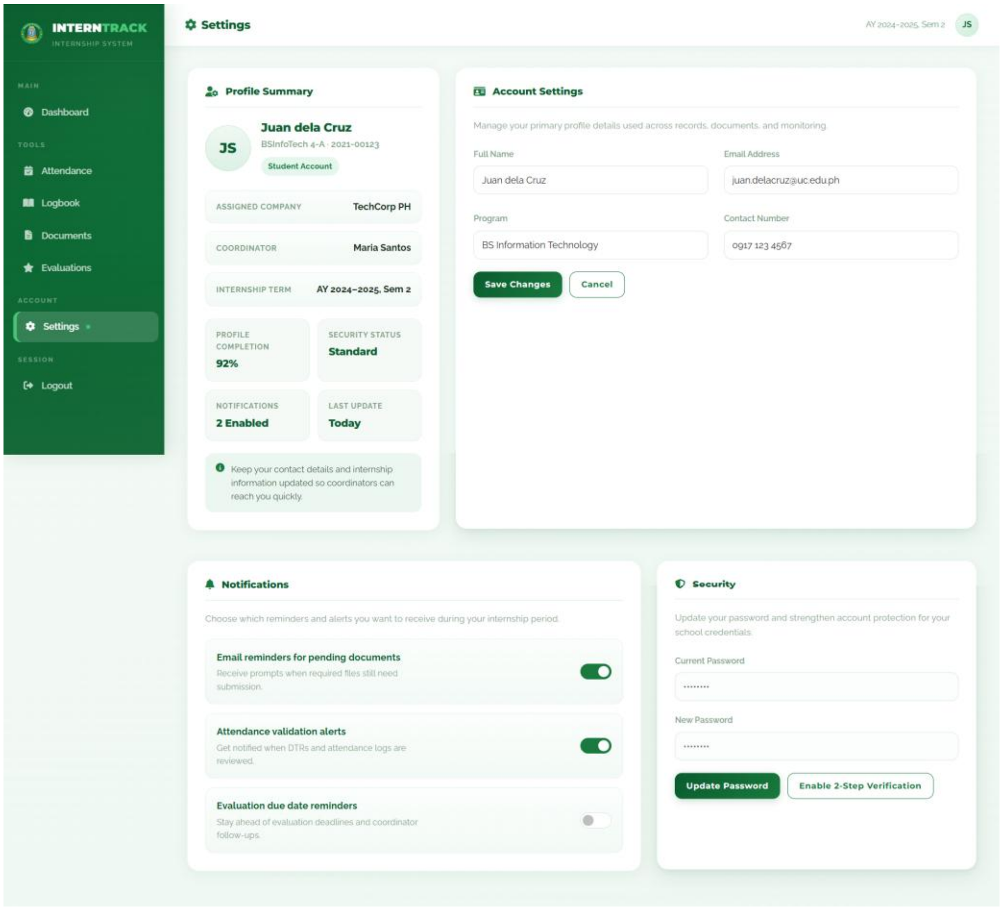

Figure 37. Student Settings Interface

The coordinator hub in INTERNTRACK serves as a centralized dashboard that provides coordinators with an overview of key internship metrics such as active interns, completion rates, at-risk students, weekly hours, and supervision records. It includes features like performance summary cards, progress graphs, supervision tables, and quick access to core modules such as approvals, reports, and logbook reviews. Overall, the interface improves efficiency and decision-making by consolidating monitoring tools into one workspace, enabling timely interventions and effective internship management.

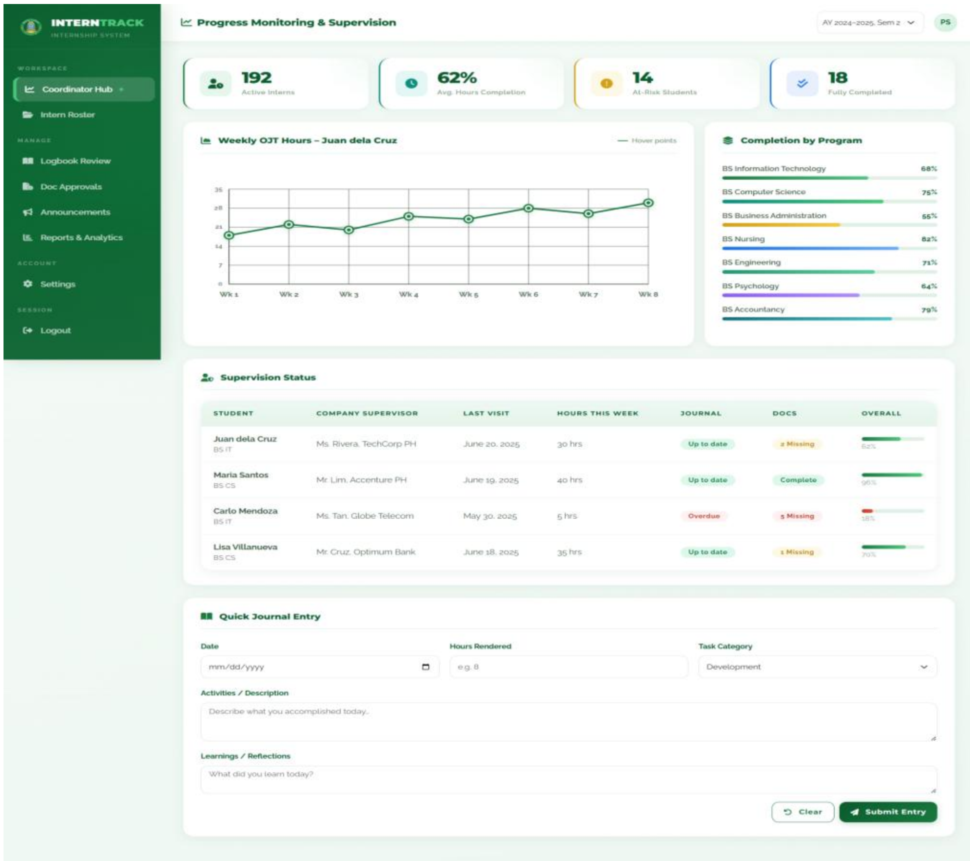

Figure 38. Coordinator Hub or Progress Monitoring and Supervision Interface

Student records and profiling interface in INTERNTRACK allows internship coordinators to search, filter and maintain student internship information including identification information, company assignment, intern hours rendered, and status, all within a single system. It features essential features such as search and filter widgets, summary cards, a well-organized intern list table, action buttons, and page controls to facilitate effective data handling. The interface enhances the usability and visibility of the platform, providing a streamlined and user-friendly way to manage student records, track progress, and make data-driven decisions.

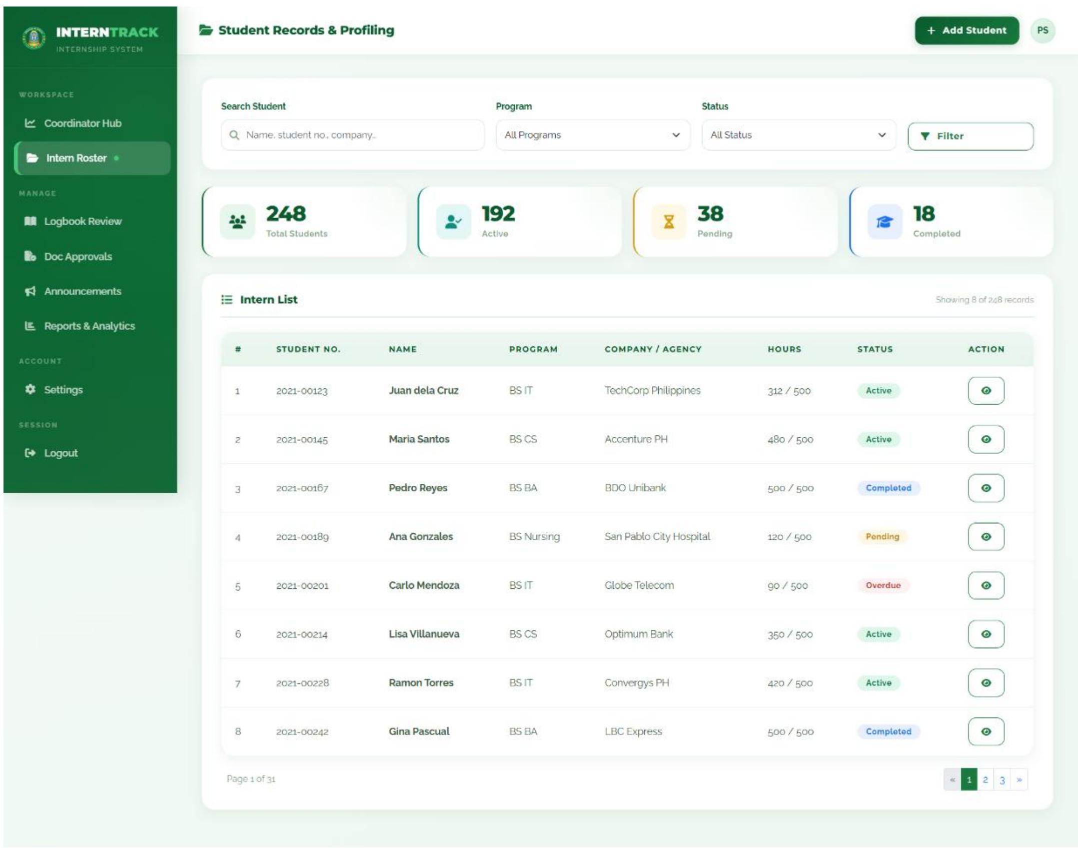

Figure 39. Coordinator Student Records and Profiling Interface

The logbook review interface on INTERNTRACK enables the coordinator to review and approve, return or provide comments on the students' logbook in one, structured environment. It contains summary cards, filters, review panels, submission details, and other action buttons like Approve, Return, and Add Comment for detailed assessment. Overall, the interface enhances the supervision and efficiency, allowing for ongoing feedback, streamlined review and improved communication between coordinators and student interns.

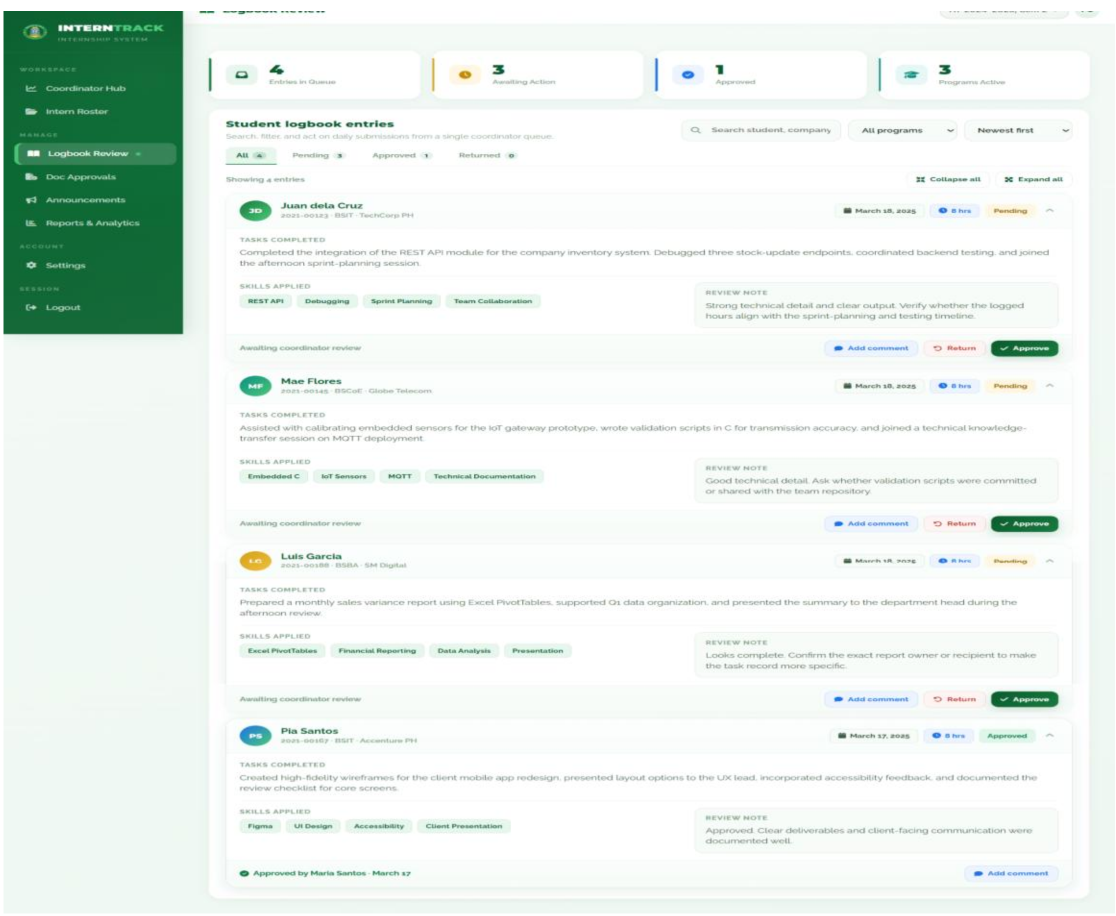

Figure 40. Coordinator Logbook Review Interface

The document approvals workspace within the INTERNTRACK shows coordinators a structured interface with a queue for student document submissions they must review and approve or return, for appropriate compliance management. It has the advantage of summary cards, search/filter functions, status tabs, requirement cards, and decision controls including Preview, Approve, and Return or Batch Processing which will help to boost efficiency. Overall, the interface improves usability and transparency, streamlines document submissions, minimizes decision delays, and facilitates centralized document validation and tracking.

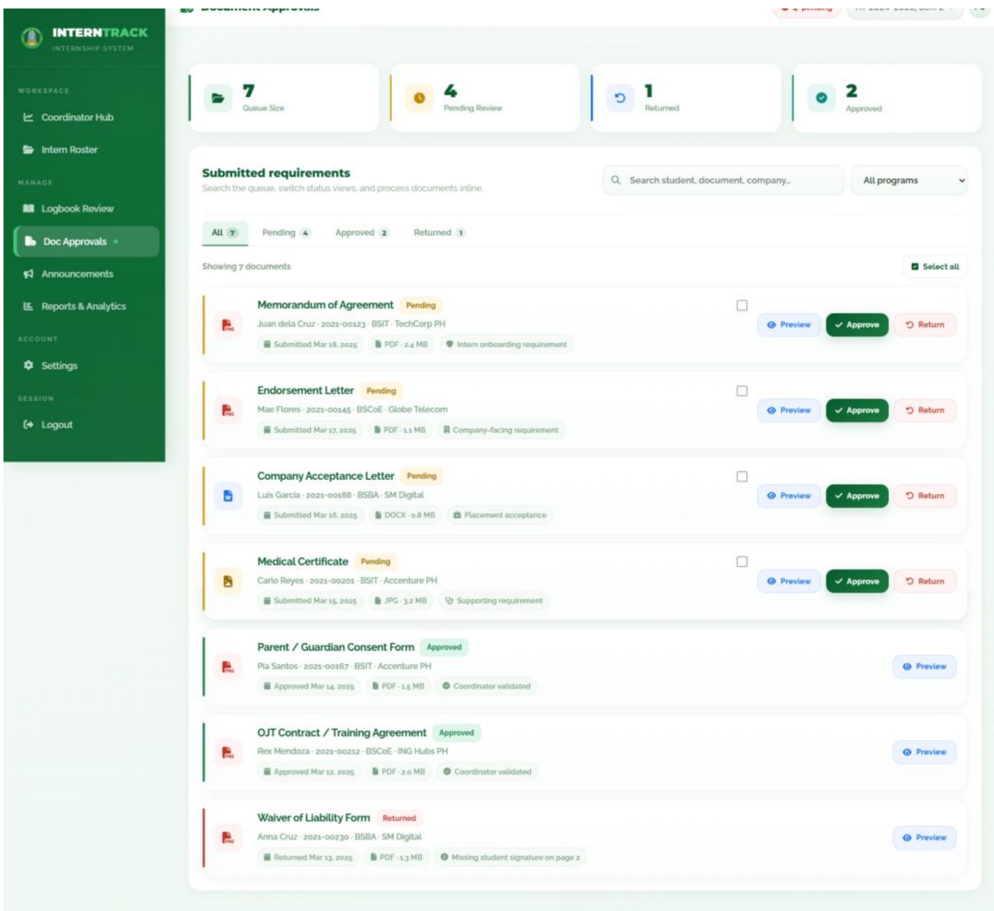

Figure 41. Coordinator Document Approvals Interface

The announcer interface of INTERNTRACK enables coordinators to produce official communications for students or groups of students in the internship system, target them and publish them. It also adds the capabilities of an announcement composition panel, an audience selector, a message field, a summary of publications, and a list of past announcements for editing and deletion. In general, the interface enhances communication effectiveness and uniformity by delivering an organized, role-based, and centralized solution for disseminating pertinent internship details.

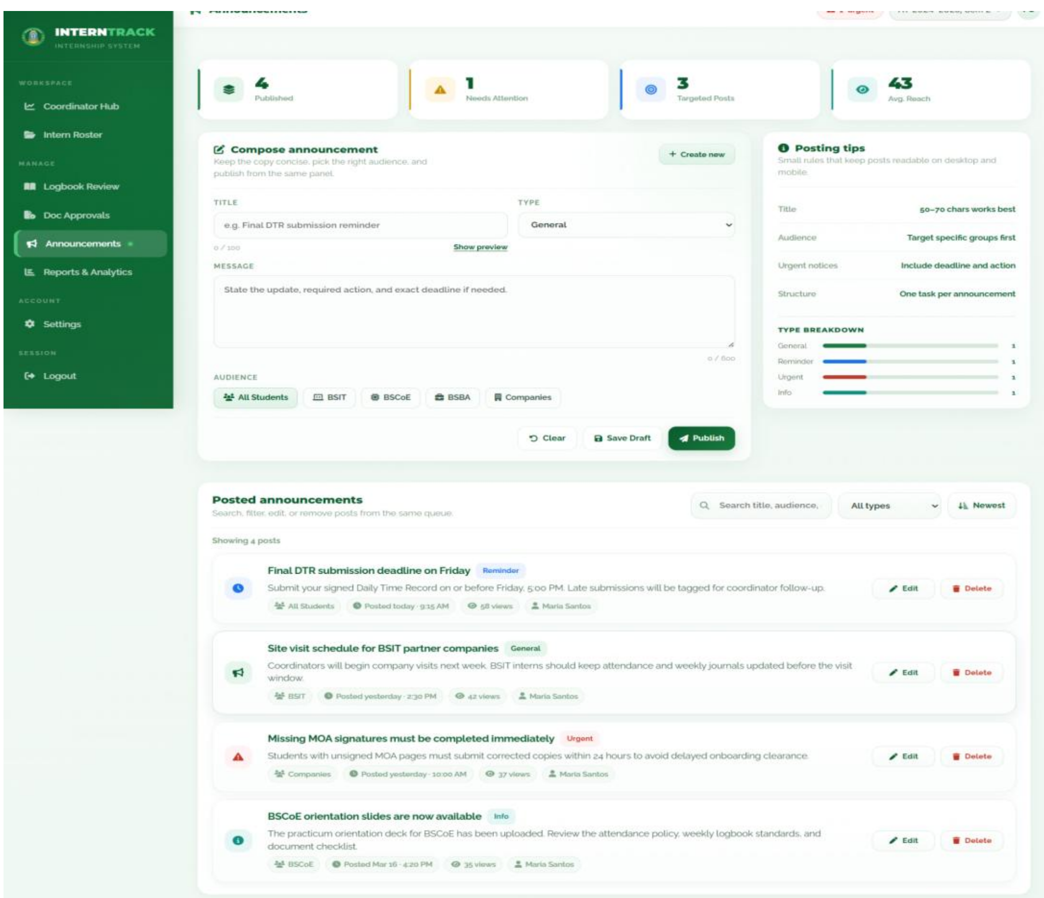

Figure 42. Coordinator Announcements Interface

The reporting and analytics interface in the INTERNTRACK allows coordinators to view internship performance, filter data and create reports about internship attendance, hours rendered, internship completion, company participation, at-risk students, and more. This also contains attributes such as summary cards, graphs, filter controls, export, and detailed tables that convert raw internship data into meaningful insights. Overall, the interface facilitates effective decision-making and supervision by offering a centralised, data driven platform to monitor and assess the performance of internship programs.

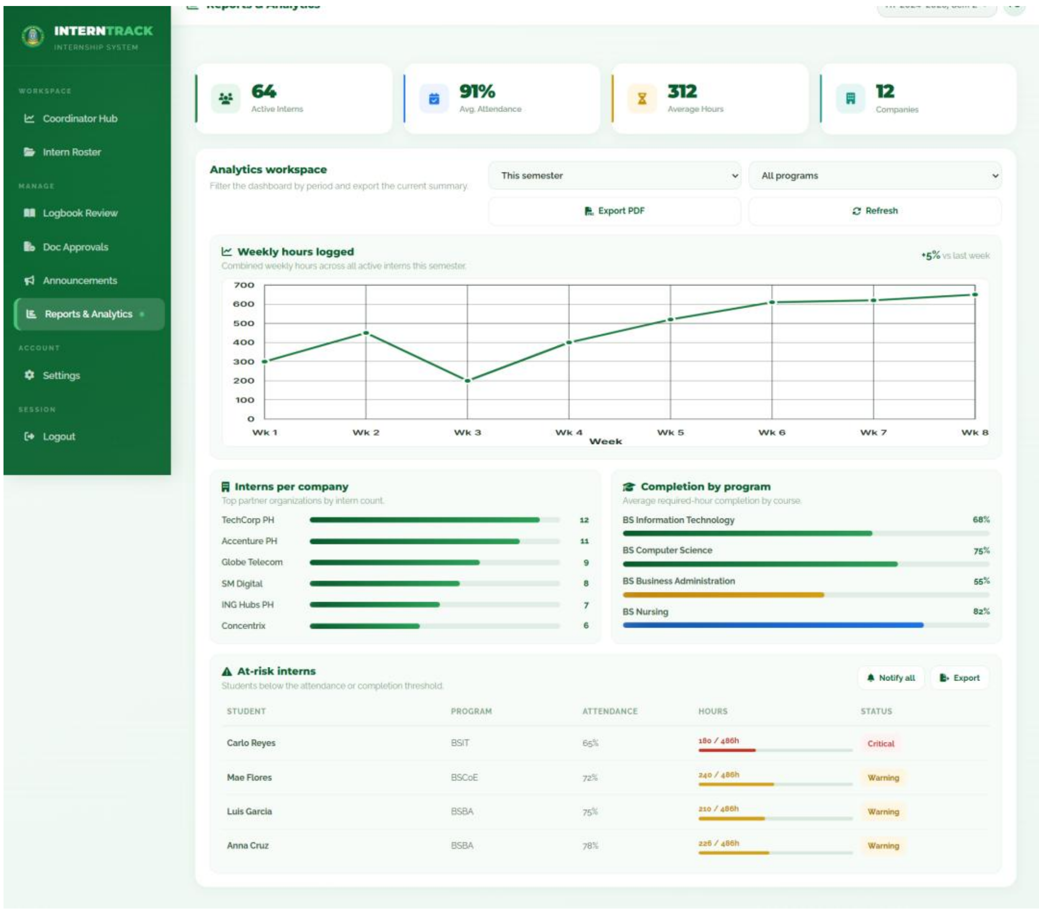

Figure 43. Coordinator Reports and Analytics Interface

The coordinator settings portal in INTERNTRACK enables coordinators to control workspace parameters, profile data, security settings, and notification preferences from a single point. It features features like profile summary, account forms, notification toggles, security controls, and defaults, digests, export formats and reminders that can be adjusted by the coordinator. In sum, it improves usability and security by creating a clear, role-specific work environment that lets users tailor workflows and control access to the system.

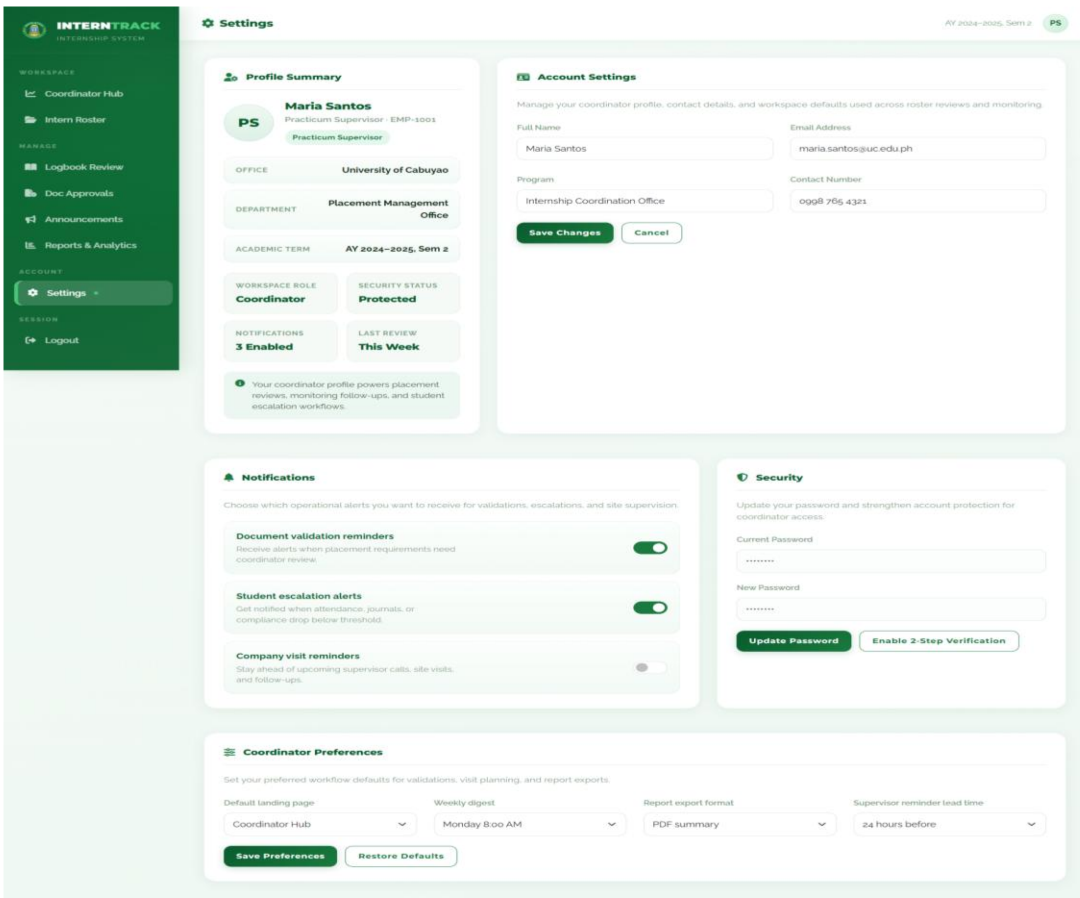

Figure 44. Coordinator Settings Interface

Ethical Considerations

This research project includes the design, development and assessment of a digital information system that has human participants and ethical compliance is important. Various stakeholders such as student intern, student internship coordinator, faculty supervisor, industry supervisor and authorized academic staff are involved in the study. It also includes requirements and testing, and assessing how the system performs. INTERNTRACK handles electronic records such as profiles, journals, attendance submission, and compliance paperwork and evaluation outcomes. The principles followed in this study are: respect for persons, beneficence, justice and compliance with the Data Privacy Act of 2012 (Republic Act No. 10173) in the Philippines. The principles will guide the recruitment of the participants, the gathering and processing of research data, and the application of protection measures for the system and the data it contains.

Voluntary Participation. This study is open to anyone interested. The researchers will ensure that there is no undue pressure, coercion and institutional influence for recruitment of student interns, faculty supervisors, internship coordinator, industry supervisors, or authorized academic staff. Each participant has the right to refuse or withdraw from the study at any time without any adverse effects to their academic, employment, professional affiliation or institutional resource status.

Informed Consent. The researchers will give a detailed description of the goals, scope, methodology, timelines, participant involvement and potential risks or benefits of the study before asking for questionnaires, interviews or evaluations of systems. A written informed consent form will be provided to each participant and they will be fully informed about what information will be collected and how it will be used in the thesis and in the system validation. Data collection will begin after participants obtain informed consent, documented.

Confidentiality of Participant Data. All personally identifiable and role-specific information will be held confidential by researchers. These will be systematically anonymised, coded or presented in aggregate form, as will direct identifiers (such as name and institutional role) to ensure that individual participants in the results cannot be identified. This will ensure that respect for people is maintained and that it is consistent with the confidentiality rules provided for by Republic Act No. 10173.

Data Security. The researcher(s) will take necessary technical and organizational measures in order to ensure the security of all data collected, especially considering that the study will be carried out with the support of digital records for organizing internships. Electronic files will be kept in password protected equipment or in limited repositories, and raw data available only to those authorized members of the research team. The data will be kept only for a duration of time as needed for analysis, validation and internal audits and will be securely disposed of according to the data management procedures established.

Handling of the E-Documents. The study will involve different types of electronic internship documentation including journals, logbooks, attendance records, endorsement letters, memoranda of agreement, and evaluation forms. These materials will be gathered, retrieved, edited and archived in line with the development of the system, the results of workflow testing, as well as research validation. Unauthorized duplication of these documents, improper transfer, storage, or use of these documents for purposes other than research is strictly prohibited.

System-Level Information Protection. Ethical responsibility in this context not only involves the treatment of the participants in a respectful way but also involves the thoughtful design and operation of INTERNTRACK. The system will enforce access control based on users' roles and provide access to information and functionalities according to specific roles of student interns, coordinators, faculty and industry supervisors, and qualified academic staff. In addition, steps will be taken to prevent unauthorized changes to the records and disclosure to third parties to maintain data integrity. This will include the use of validation controls and controlled testing facilities during development and evaluation process.

Non-Maleficence. The researchers are dedicated to designing and executing the study in a way that will not cause harm; they adhere to the principle of beneficence. The procedures used will not result in any risk to participant physical, psychological, academic or professional interests beyond their participation in the research and system evaluation. The study will also not employ manipulative techniques, invasive questions or inappropriate access to internship information that might give rise to discrimination of participants.

Equity and Impartiality. Justice requires that everyone who is eligible for the research, be treated fairly and without discrimination. Researchers will not give preferential or discriminatory treatment based on academic programme, prestige of academic institution, job position, or personal connections with the research team. The methods of recruitment, distribution of research instruments and analysis of the results will be uniform so that the procedures used in the study are uniform and ethical.

Institutional Authorization. The researchers will secure the necessary approvals from the necessary office or academic institution at the University of Cabuyao before formal data collection and system validation can be undertaken. Participant access, institutional procedures and relevant records will be limited to that which is allowed by the university and compliant with internal research and data governance procedures. This is necessary to assure institutional accountability, respect administrative procedures and legal requirements for handling institutional information.

Limited Usage of the Data. Data collected as a result of the needs assessment, system testing and performance evaluation will only be used to fulfil the aims of this thesis. The researchers agree and guarantee that they will not disclose, reuse or commercially use any responses, scores, records or documents made from the study to a third party without further consent if required. After the research is complete and the designated retention period has passed, all sensitive records will be destroyed or deleted permanently to ensure that the records are not used by any unauthorized parties in the future.

To sum up, the moral standards that are highlighted in this research are human subject and computer record protection. The researchers declare their adherence to the principles of respect for persons, beneficence and justice, as well as institutional policies and the Data Privacy Act of 2012. This will help to ensure that the research is done in a legal, ethical and safe manner, which will allow for the responsible development and assessment of INTERNTRACK and the protection of the rights, welfare and privacy of all parties involved.

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